

Volleyball Player-Impact Metrics

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2025

A thesis submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Electrical (Computer or Telecommunications) Engineering at The University of Newcastle, Australia.



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Abstract

This project investigated regression-based player evaluation techniques in application to volleyball. Ultimately, the aim is to advance volleyball analytics to a mature state beyond the box-score. Building upon a review of current volleyball-specific player evaluation techniques, this work explores a variety of areas for iteration in regard to regression-based methodologies. The first is a widely used approach comparing point-outcomes against player court-presence (who is on the court). The second instead uses touch-presence (who touched the ball) to predict the team's chance of scoring a point. The values returned by these models were used to compute a Points Added for each player. A variety of augmentations were applied to these models with the goal of improving their performance. Model performance was analysed in the following dimensions: Predicting team success, cross-validation ability, biasing, and data granularity. The study revealed that adjusting for the serve and differentiating between front-court and back-court had the strongest impacts. On the other hand, touch-based models had some predictive power but were generally much more unstable and context-dependent. The augmentations that most improved the touch-presence model were those incorporating the opponent, either by their touches or court-presence. In summary, while the touch-based approach offered some interesting insights, it proved too unstable for practical evaluation. However, the popular court-presence approach may suffer from similar issues. Despite this, a range of strategies and transformations have been tested to improve these models in future applications of regression-based modelling and the continued development of volleyball analytics.

Key Contributions

- Comparison between two leading player-evaluation techniques in volleyball (presence-based and touch-based regression), especially regarding prediction ability and data characteristics.
- Exploration of a range of design matrix augmentations and their effects on regression outputs.
- Iteration to a skill-touch based model for player evaluation, featuring additional data granularity and a predictive accuracy comparable to traditional court-presence models.

Acknowledgements

Thank you to my supervisors of course, Lawrence Ong and Ian Renner. Fantastic.

Thank you to the various members of the sports analytics scientific community for going above and beyond in their assistance.

Thank you to all my friends and family who suffered through my ramblings.

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Chapter 1

Introduction

1.1 History and Motivation

Sports analytics is a storied and continually growing area. A zeitgeist of the field is the novel *Moneyball* [2], following the application of statistics to objectively measure player performance in the interest of identifying undervalued players. Sports analytics attempts to provide a quantitative and data-backed approach to decision-making [3].

Major sports such as baseball, basketball, hockey, and soccer have all developed major analytics industries [4]. Many franchises employ dedicated statistics departments. Most NBA teams support more than 20 analytics employees to inform strategy and management decisions. As an example, Daryl Morey's Houston Rockets were early on an NBA trend toward more 3-point attempts. The number of attempts has increased by 74% in the last 20 years [5]. Pertinently, analytics has applications in the evaluation of specific players. This can be used to inform line-up decisions [6], playing time, trade evaluations, contract offers, and strategic analysis. It asks the question: How valuable can one player be? What makes them valuable?

Volleyball has lagged behind this analytics boom despite its popularity. In 2023, the Nebraska-Omaha matchup set the record for attendance at a women's sporting event with 92,003 attendees [7]. This was broadcast open-air in a stadium typically reserved for football games. Moreover, volleyball is highly suitable for quantitative analysis, due to the high number of possessions, discrete actions, and stable playing environment [8]. As a point of comparison, the Federation Internationale de Volleyball (FIVB) is the governing body for international volleyball. They keep basic 'counting' stats. This includes totals such as hits, sets, receives, and serves. In contrast, the National Basketball Association (NBA) lists several more developed statistics such as Player Impact Estimate, offensive rating, and rebounding percentage [9].

There have been steps in developing player evaluations for volleyball. Science Untangled [10] is an organisation specialising in this field. They use ball and player tracking to analyse

trends such as attack distribution, evaluate player decision-making, or identifying systems. Furthermore, there are a suite of scouting programs using computer vision or hand tracking for highly detailed game information [11]. These provide an excellent foundation for future analysis. Additionally, volleyball has received a hefty exploration on the effects of certain skills. In regards to player evaluation specifically, a mix of action-based and presence-based approaches were taken.

While other sports feature dozens of analytical studies, volleyball remains relatively unexplored with only five papers identified on the topic - only three of which introduce novel concepts. The public presence and language of volleyball statistics is still primitive. The sport is a prime subject for analytical studies. As a result, this area is a highly promising and just now burgeoning field. The aforementioned papers were published in 2024, 2024, 2023, 2022, and 2017. Continued iteration, experimentation, and formulation will further push volleyball analytics to a mature state.

1.2 Thesis Objectives and Parameters

This project aims to investigate the statistical properties of player-impact models in volleyball. The goal is to improve understanding in the field of volleyball analytics and to identify the key factors influencing statistical models as they evaluate player ability.

Doing so assumes that player impact is a latent ability constant across games. Furthermore, that this is measurable from player touches and rally outcomes. The player evaluations will be conducted per-league, and attempts will not be made to project across multiple leagues. Whilst some contextual factors such as home-court will be taken into account, physiological effects such as fatigue, mindset, or injury will be ignored.

The process for conducting this study will begin with a literature review on the current landscape of both general sports analytics and volleyball specifically. Subsequently, suitable data sources must be found, noting the features available for analysis in these data sets. This will be consolidated with other sports analytics, identifying any key areas or characteristics of volleyball which are ideal for analysis. Finally, models will be built to evaluate the impact of players using a variety of information inputs and techniques. The created metrics will be compared to assess their accuracy, reliability, and any properties associated with the model.

The success of this study will be based upon the accuracy, robustness, and applicability of the models created. A perfect model should be able to recreate the outcome of games based upon the players involved. The predictive power will also be consolidated with a common technique of cross-validation. Furthermore, the results can be compared to external indicators such as player ELO, awards, public sentiment, or team success.

By exploring how different modelling techniques and sport-specific factors influence player evaluation, this work aims to establish a foundation for future volleyball analytics research.

A reliable and objective model for player evaluation will contribute to volleyball's sporting industry. Coaches, managers, and fans alike may use it as a reference point for player discussion and analysis. Furthermore, it will give insight into the nature of volleyball: The degree to which it is a player-centric or team-centric game; the importance of positions; and the effect of key skills.

1.3 Thesis Outline

1.3.1 Introduction

This chapter provides an introduction to the paper. The motivation, scope, and research goals.

1.3.2 Background

This chapter gives insight into the surrounding information necessary to understand for the paper. Particularly, the structure of the sport of volleyball, and some commonly used statistical terms.

1.3.3 Literature Review

This chapter gives insight into modern methods of player evaluation, a review of statistical studies into volleyball, and the methodologies that have previously been applied to volleyball for player impact. It should give an idea of the landscape of the field, and the novel opportunities available.

1.3.4 Regression Methodology

This chapter details how the regressions will be conducted - from data acquisition, matrix construction, and finally to computation.

1.3.5 Model Augmentations

This chapter details a range of matrix augmentations designed to improve results. The reasoning, implementation, and hypotheses for each is detailed.

1.3.6 Model Evaluation Methodology

This chapter details how models will be compared in terms of predictive accuracy and the criteria separating models.

1.3.7 Results

This chapter outlines the results of these algorithms, as well as how they compare with initial hypotheses.

1.3.8 Discussion

This chapter engages in discussion of the results. This includes general trends, shortcomings, and observations of the nature of the data.

1.3.9 Conclusion

This chapter makes conclusions based on the works of this paper and outlines opportunities for future work.

Chapter 2

Background

2.1 Structure of Volleyball

Volleyball is a six-a-side, high-tempo sport. Like most net sports, teams play for a point - in this case by landing the ball in the opponent's court or otherwise inducing an error. Unique to volleyball is a limit of 3 touches per side, introducing highly modular elements and skills. Most sources list 6 key skills based on these touches. They are: serving, receiving, setting, hitting, blocking, and digging [12]. Performing these skills has a variety of names such as a 'touch', 'action', or 'skill outcome'.

Another distinct aspect of volleyball is the rotation. Three players occupy the front-court, and three players the back-court. Back-court players are not permitted to block or make front-court attacks. Players rotate clockwise as the game progresses.

A general rally structure is as follows:

- **Serve:** From behind the baseline by the serving team.
- **Reception:** By the receiving team, to their team's designated setter.
- **Set:** Setting the ball to an optimal location for a hitter.
- **Attack:** Hit in an attempt to score a point.
- **Block:** Block at the net by the opposing team's front-court players.
- **Dig:** Dig by the opposition, directed towards their team's designated setter.
- The set, attack, dig, chain continues.

Within this system, there are 5 different roles, each specialising in a few of the key skills:

- **Outside Hitter:** Hitters and passers; hit and block from the left side.
- **Opposite Hitter:** Hitters and blockers; hit and block from the right side.

- **Middle Blocker:** Hitters and blockers; hit and block the middle, as well as helping Outside and Opposite to block at the sides. Typically, they are the tallest player on the team and rarely participate in reception or digging.
- **Libero:** Passers and defensive specialists; have special rules allowing for additional substitutions, usually substituting for the weakest passers (middles).
- **Setter:** Responsible for setting on the second touch.

The usual playing position of each role is visualised in Figure 2.1. The court is split into 6 sections which are typically referenced for positioning purposes and data recording.

Each game is split into a best of five sets. Each set is played to 25 points, also called rallies. Each point has distinct phases of play often labelled as possessions or complexes. These are defined by the consecutive touches by a team whilst it is on their side. Complex 0 refers to the serve. Complex I refers to the receiving team. This involves the reception, set, and attack. Complex II refers to the serving team. This includes the serve, block, defence, and counterattack. Complex III is any continuing rally thereafter.

2.1.1 Data Recording

The most prevalent data recording software is Data Volley. This uses manual entry to document all touches in a match. Many organisations and specific teams use this software [13]. The results are stored in .dvw files. For each game, the rotation, score, and substitution are all recorded. Furthermore, each action has an associated rating and location. The evaluation is typically on an ordinal 1 - 6 Likert scale represented by a symbol [11]. For example, S# refers to a service ace, and S= to a service error. Whilst Data Volley is not universally adopted, many alternatives record similar information.

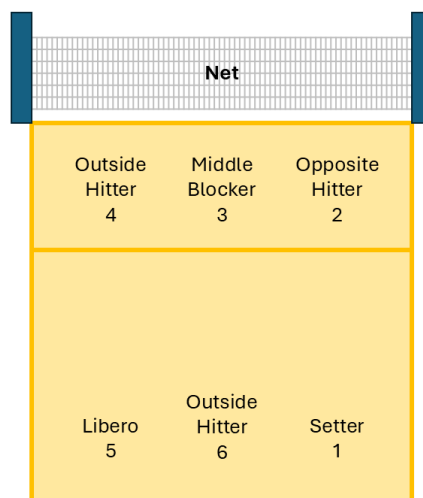


Figure 2.1: The positions of volleyball players in a typical playing rotation.

Types of Statistics

The most common statistics viewed by the general population are known as box-score statistics (also known as counting stats). These are a record of individual player actions. In soccer, this may be goals, assists, tackles, etc. In the case of volleyball, typically 6 actions are recorded: Serves, receives, sets, hits, digs, and blocks. These are generally recorded as totals of attempts and successes.

Play-by-play statistics provide a far more comprehensive review of the match. They keep an ordered record of the game, as well as other highly valuable information. For example, players on the court, score, or possessions. Play-by-play data may also include locations and more granular actions that aren't typically tracked [14]. These play-by-play stats demand a much higher workload from the scouter.

2.2 Common Analytical Tools

2.2.1 Bayesian Methods

Bayes' theorem is a simple equation relating conditional probabilities [15]. It gives the likelihood that some condition is true, given an observed result and a 'prior'. The formula is given in (2.1).

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (2.1)$$

The application of Bayes' theorem relies heavily on the prior $P(A)$ and returns a posterior $P(A|B)$. Unlike frequentist methods such as regression, the posterior is a distribution with both expected value and variance.

The prior is the key parameter in the design of Bayesian models. A common approach to setting a prior is through random effects. The concept behind random effects is that each observation is not an isolated case but instead drawn from a population distribution.

For example, suppose hitting percentage θ is a random variable drawn from normal distribution $\theta \sim \mathcal{N}(\mu, \sigma^2)$. Here, μ and σ^2 refer to the mean and variance of population hitting percentage. Rather than calculating each player's hitting percentage independently, the random effects model assumes each player's θ_i is a realisation from this distribution. This introduces a degree of pooling, where extreme values with small samples are shrunk towards the mean.

2.2.2 Regression Methods

Linear Regression: When interpreting the relationship between variables, regression is an intuitive and bespoke approach [15]. Given observed relationships between output and input, a linear model can be built:

$$\hat{Y} = \beta_0 + \beta_1 X + \varepsilon$$

Here, Y is the response variable, X the predictor variable, β_0 the intercept, and β_1 the gradient representing how Y changes with X . Most data does not exhibit a perfectly linear relationship, and a straight line will not cross through all data points. Rather, a trend line is produced where $\hat{Y}$ is the expected value $E[Y]$ for that X value. The parameters β_0 and β_1 are tuned to minimise the Residual Sum of Squares (RSS). The cost function attempting to be minimised is given in (2.2).

$$Q = RSS = \sum_{i=1}^n (Y_i - E[Y_i])^2 \quad (2.2)$$

Here, Y_i refers to the outcome of observation i , with predictors X_i , The $E[Y_i]$ term refers to the value predicted by the model.

This method of producing a trend line follows a deterministic maximum likelihood approach and assumes normally distributed error. The trend line can be used for interpolation, extrapolation, and identification of variable relationships.

Logistic Regression

Typically, regression methods are unsuitable for classification. However, in the case of binary outcomes such as win or loss, the outcomes can be assigned as 1 and 0. Directly applying regression to such a model is inaccurate, and as such taking a logistic approach is more suitable. The formula for logistic regression appears in (2.3).

$$P(Y = 1|X) = \frac{1}{1 + e^{-\hat{Y}}} = \frac{1}{1 + e^{-\beta_0 + \beta_1 X}} \quad (2.3)$$

Where $P(Y = 1|X)$ indicates the probability of outcome $Y = 1$ given the predictor X can be modelled as the log function of some expected outcome $\hat{Y}$. This expected $\hat{Y}$ is usually represented as a linear function of X ; $\hat{Y} = \beta_0 + \beta_1 X$

Logistic regression is often applied in sports analytics to gauge the importance of skills. It is an interpretable method for assessing whether certain actions are related to winning a point. Using logistic regression, each variable has a coefficient for how it alters the log-odds of victory. Compared to other classification models, logistic regression can use binary variables effectively.

Chapter 3

Literature Review

3.1 Introduction

As in many other industries, data analysis has become an essential part of management in sports. This may lie in forecasting wins, informing strategic decisions or player valuations, and making business decisions [16].

Volleyball is a mix of many fields. Like racket sports, it features a net and begins with a serve, with the objective of playing a ball the opponent cannot return. However, it is a team sport with clear features and holds discrete information due to the nature of the 3-touch limit. And whilst the opposition is separated, play is continuous once the rally begins, with complex interactions between skills. Given its multidimensional nature, volleyball lends itself to a broad spectrum of analytical approaches

3.2 Plus-Minus Models

Plus-minus (represented by PM hereafter) is a popular and incredibly valuable statistic in sports analysis. The formulation is the difference in team score while a player is on the court. It is the preeminent *top-down* statistic [8]. Bottom-up statistics are those calculated based upon player actions. In contrast, top-down ratings use the performance of the team and credit this performance based on the presence of players in the match. PM is particularly popular in basketball and hockey due to the high number of possession and player swaps. This section focuses on basketball, where PM metrics are most popular and most extensively utilised.

The calculation of PM is simply how the score changes while a player is in the game. If a player were to enter the game whilst the score is 30-30, and exit the game when the score is 45-40, that player would have a plus-minus of +5.

By referencing change in score, PM provides a holistic measure of player impact. Bottom-up statistics are restricted by the measurable actions in the game. On the other hand, PM

is influenced by various 'intangible' factors, such as communication, positioning, defence, or even change in tactics. Any facet that may influence the game will be reflected in the change of score.

Despite the low scoring and substitution rate in soccer, there have been attempts to apply PM using proxy statistics. Other parameters - such as shots on goal - may be used instead of the pure score differential [17].

However, there are key shortcomings in PM. Firstly, it does not explain how a player influences the game. Secondly, it is incredibly noisy as it only captures marginal impact. That is, it does not account for the other players or events on the court. A player F who always shares the court with elite teammates will share in their impacts on the score. This may bias the PM of player F much higher than their 'true' skill would imply. To solve this, there are many techniques, interpretations, and applications of plus-minus to bring context to this information.

3.2.1 Plus-Minus Per Possession

Raw PM can be divided by the number of possessions a player is involved in. Possessions are the number of times the team controls the ball, and thus has an attempt to score. It acts as a highly stable rate. In basketball, most PM is captured per 100 possessions to more intuitively represent the data. This will be shortened to PM/100 for conciseness. A player involved in 50 possessions with a PM of -2, will have a PM/100 of -4.

To better isolate this player's impact, the team's PM/100 whilst they are off the court is also taken. The difference between the on and off-court PM/100 for a player is referred to as an On/Off split. For example, a player may have an on-court PM/100 of -2, indicating the player's team performs poorly whilst they play. However, their off-court PM/100 possessions may be -10, significantly worse. We may interpret the player as making a positive impact, and the team is simply non-competitive. Their On/Off per 100 is said to be +8 in this case.

Furthermore, this can be augmented for offensive and defensive possessions, to separate the net impact into its offensive and defensive components. This is sometimes referred to as a player's *offensive rating* and *defensive rating* [18].

However, PM/100 and On/Off per 100 remain marginal effects. Starters typically have inflated On/Off per 100, as they start and come off with the best players on the team.

3.2.2 Adjusted Plus-Minus

To identify the partial effects of a player, the other players on the floor must be controlled for. Rosenbaum is credited with the synthesis of a method known as adjusted plus-minus (APM) [19]. Each period of the game where all players remain the same (no substitutions) is treated as an observation. These periods will be referred to as a 'split'.

The player matrix has columns relevant to each player, and rows relevant to each split. The home players involved in a split are represented by +1, and visiting players by -1. Those not playing are a 0. Each split has an appropriate 'marginal' indicating the score difference in points per possession.

Thus, the marginal can be treated as a linear function of the players in the game. The model thus assigns an APM value to each player. The marginal should be equal to the APM of the home players subtracted by the APM of the visiting players.

The formulation is given in (3.1). A linear regression can be run using these matrices.

$$Y_i = \beta_0 + \sum_{j=1}^K \beta_j X_{ij} + \varepsilon \quad (3.1)$$

Here β_0 refers to home court advantage and ε the error term. The player matrix entry X_{ij} indicates whether player j was on the court during split i . The Y_i term refers to the resultant marginal in this split. Each β_j is the APM coefficient given to player j and indicates the difference in APM between them and a replacement player. The K value simply refers to the number of players.

Replacement players are non-starters or bench players. The concept of a replacement-level player is very important in these metrics. It is necessary to note the average player performs below the average standard of play. For example, a team of 15 will have 5 starters in a basketball game. These strongest 5 play the most minutes. Whilst the 7th or 8th best player may be average amongst all players, they perform below the average of all minutes played. These replacement players typically have a negative impact. In Rosenbaum's case, it was those who did not pass a playing time threshold. These players do not appear in the regression at all and are typically assigned some negative value.

Other contextual factors may be included. For example, observations may be adjusted based on the closeness of score. Typically, the larger a lead, the worse the leading team tends to perform as they relax.

It is important to note that all variations of APM treat each player as an additive element. This is inaccurate due to various interaction effects between teammates, opponents, or strategies. A player traded between teams will rarely retain the same APM values.

Whilst APM has been hugely foundational for basketball analysis, it is extremely noisy and can produce extreme values. A key issue is the extreme degree of multicollinearity, where most players tend to play with the same players. For example, a common pairing of two players may consistently produce a marginal of +2. The purely linear APM model may assign a +20 to player 1 and -18 to player 2. A common method of splitting this credit is by the use of regularisation.

3.2.3 Regularised Adjusted Plus-Minus

A highly popular method of stabilising APM is by the use of regularisation terms, especially ridge regression. This is referred to as Regularised Adjusted Plus-Minus (RAPM) [8].

Linear regression typically attempts to minimise the difference between predicted and actual results. The cost function is the difference squared. Regularisation adds an additional cost element based upon the magnitude of generated coefficients. This is also referred to as shrinkage since the magnitude of coefficients is penalised and thus pulled towards 0. Ridge regression applies an L2 penalty, using the square of parameter values (β^2), whilst lasso applies an L1 penalty using the absolute values ($|\beta|$). The ridge method splits value amongst collinear terms, as the squares of large numbers add heavy costs and so two moderate numbers are preferred. Lasso on the other hand can perform feature selection, as certain coefficients are shrunk to exactly 0 as best fits the cost function. Applying ridge regression to APM modifies the cost function as in (3.2).

$$\sum_i \left(Y_i - \beta_0 - \sum_{j=1}^n \beta_j X_{ij} \right)^2 + \lambda \sum_{j=0}^n (\beta_j)^2 \quad (3.2)$$

Here:

- Y_i is the true marginal observed in split i .
- β_0 is the intercept due to home court advantage.
- β_j is the APM coefficient for player j .
- X_{ij} indicates the presence of player j in split i .
- n is the total number of players.
- λ refers to the regularisation parameter controlling the degree of shrinkage.

This accounts for multicollinearity by splitting the credit for the marginal between players. Lasso is generally not employed as it assigns most players to a 0 value [20]. Ridge regression performs a more equal split between active players. RAPM performs better in almost all circumstances as compared to APM [21].

The regularisation parameter λ is variable. Cross-validation is a common technique used to determine suitable values.

This regularised approach forms the foundation of modern player impact modelling and directly inspires the methods used in this project.

3.2.4 Statistical Plus-Minus

An extension of these adjusted concepts is through Statistical Plus-Minus (SPM), an attempt to stabilise these metrics and give weights to typical box-score statistics [19]. In

essence, each player's RAPM is used as the outcome Y . The player's box-score statistics (points, rebounds, assists, blocks, etc) are used as predictors. A regression is ran to predict RAPM based upon player game statistics. This is very flexible, and a variety of game statistics can be used for this process. Perhaps the most famous example is Box Plus-Minus (BPM) by Daniel Myers, which uses box-score stats per 100 possessions [22] as the predictors.

This produces a list of coefficients for each statistic used as a predictor (E.g. A player's 4 assists per 100 possessions are relative to a $+0.8$ for their RAPM score). Multiplying a player's statistics by these coefficients produces their SPM. This lies at the heart of SPM models. However, modifications can be made. Daniel Myers' BPM model will be used as an example [22]. Myers makes the following additions: Adjusting coefficients based upon player role; adjusting shooting based on total team performance; using priors based upon team performance and player minutes; and ensuring the sum of BPM equates to the team's rating.

The use of SPM models is typically more stable than RAPM or APM, as game statistics tend to be far more consistent than score differential. Rosenbaum produced a combined metric, weighting raw RAPM and SPM. Using cross-validation, this method was found to be a more accurate prediction of final RAPM value than just RAPM [23].

3.3 Volleyball Studies of Note

3.3.1 Skill Evaluation Papers

Given the discrete skills in volleyball, many papers explore which skills are most valuable. Silva conducts a literature review for volleyball match analysis [24]. They produce tables for skills, position, and match phase - both broadly and predictively. Both the sources from this paper, and a conglomeration of others have been summarised in Table 3.1.

A majority of these papers use graded actions as part of the analysis. Typically, volleyball data records both the skill and a quality evaluation. The importance of the skill is then evaluated with correlation, most commonly the Chi-Square Test of Independence. It is a measure of how well two categorical variables are related. For example, to compare the link between pass quality and match-set outcome.

Table 3.1: Studies of volleyball skill analysis and match analysis

Author	Statistic	Findings
Palao <i>et al.</i> [25]	Chi-Square Test and likelihood ratio	Team level is most differentiated by spiking and blocking.
Araújo <i>et al.</i> [26]	Chi-Square Test	A notable relationship exists between blocking system and blocking effectiveness. Blocking effectiveness showed an independent relationship to set outcome.
Liro & González [27]	Chi-Square Test and Pearson	The libero is highly influential in serve reception. There is no significant difference in reception between power service and jump service.
Monteiro <i>et al.</i> [28]	Chi-Square Test	Hitting efficiency has a significant relationship with set outcome. A similar outcome was not found for digging efficiency
Laios & Kountouris [29]	Chi-Square Test	The serving team is generally at a disadvantage. Rotation line-ups significantly alter rally outcome
Eom & Schutz [30]	Log-Linear Analysis	Identifies relationships between sequential skills. The Pass x Set and Set x Spike relationships were extremely stable. The Pass x Spike association is significant, but far weaker. The major differentiating factor between hitters is their ability to convert on the most common 'mediocre' sets, rather than poor or perfect sets.

Continued on next page

Author	Statistic	Findings
Marcelino <i>et al.</i> [31]	Pearson correlation coefficients	Spiking performance (relative only) is the best indicator of success. Block points per game is also a strong predictor.
João <i>et al.</i> [32]	Discriminant function analysis	Identifies sex differences: Women tend to have longer rallies and thus defensive skills are stronger discriminators. Men's volleyball is more associated with terminal actions such as service errors.
Patsiaouras <i>et al.</i> [33]	Kruskal-Wallis nonparametric test and Mann-Whitney U test	Strong and weak teams can be differentiated by the number of "attack errors following bad receptions".
Zetou <i>et al.</i> [34]	Discriminant function analysis	In Complex II overpasses, aces, and counterattacks all had high loadings for point outcome. Neither block nor dig were strong discriminators.
Durković <i>et al.</i> [35]	Discriminative analysis	Data when the setter is in the back row is more useful for predicting wins due to greater offensive potential. When the setter is front row, reception, defense, and counterattack become more important. Attack and service remain universally important.
Nikos & Elissavet [36]	Chi-Square Test, ANOVA, Multinomial logistic regression	First and second tempo sets strongly increase the chance of gaining a point directly from an attack
Sanghvi <i>et al.</i> [37]	Machine learning	Player aggregates outperformed team aggregates for win prediction.
Florence <i>et al.</i> [38]	Markov chain modelling	Finds little difference (0.5% win percentage) between 4, 3, and 2 point passes on rally outcome.

Collectively, these studies establish an empirical basis for the relative importance of volleyball skills. Terminal actions such as attacking, blocking, and serving consistently show the strongest relationships with match outcomes, while reception quality strongly predicts subsequent setting and attack success. These relationships provide a valuable foundation for interpreting the results of player impact models.

These papers produce a formidable backbone of the deciding factors for volleyball matches. This includes the importance of terminal actions such as attacking, blocking, or errors. Furthermore, there are strong correlations between subsequent skills such as reception quality to setting. These relationships will act as a valuable foundation for interpreting the results of the final player impact models.

3.4 Papers Attempting to Evaluate Volleyball Player Impact

Five studies were identified as attempting to quantify individual player impact in volleyball. The following section reviews these papers and their approaches to player evaluation. Approaches taken include Bayesian models, division of credit, Markov chains, and adjusted plus-minus frameworks. These papers represent the current state of quantitative volleyball evaluation. The Technical Individual Contribution Coefficient developed through the work of López-Serrano *et al.* [39] is notable. However, it will be omitted due to a lack of relationship with game outcome.

3.4.1 Points Above Average Per Set - Bayesian Logistic Model

Each possession in volleyball sustains a maximum of 3 touches. Fellingham attempts to use touch-presence to evaluate player effects on possession outcome [1]. The definition of impact in this case is how the probability of success changes when a player is inserted into a team composed of 'average' players. A Bayesian hierarchical logistic model is used to estimate the contribution of players per position.

The model uses possession result as the outcome: +1 for point scored; 0 for continuation; and -1 for point loss. Athlete touches act as inputs. Hence, the model measures how a player's touch contributes to the log-odds of a positive or negative outcome.

For each ball possession i , the result r_i is modelled as a Bernoulli outcome with probability of success p_i as captured in (3.3). Separate models are trained for the positive result +1 and negative result -1.

$$r_i \sim \text{Bernoulli}(p_i) \tag{3.3}$$

The p_i term is treated as a function of player touches as defined in (3.4).

$$\begin{aligned}
\text{logit}(p_i) = & \beta_0 + \beta_{\text{Set}_{[\text{setterid}[i]]}} \times I_{\text{setterid}[i]} + \\
& \beta_{\text{Oh}_{[\text{ohid1}[i]]}} \times I_{\text{ohid1}[i]} + \beta_{\text{Oh}_{[\text{ohid2}[i]]}} \times I_{\text{ohid2}[i]} + \\
& \beta_{\text{Mb}_{[\text{mbid1}[i]]}} \times I_{\text{mbid1}[i]} + \beta_{\text{Mb}_{[\text{mbid2}[i]]}} \times I_{\text{mbid2}[i]} + \\
& \beta_{\text{Opp}_{[\text{Oppid}[i]]}} \times I_{\text{Oppid}[i]} + \beta_{\text{Lib}_{[\text{libid}[i]]}} \times I_{\text{libid}[i]} + \alpha_{\text{Team}_{\text{teamid}[i]}} \quad (3.4)
\end{aligned}$$

Where:

- $\text{logit}(p_i)$ refers to the log-odds probability p_i of point outcome r_i for possession i . This is defined as $\log \frac{p_i}{1-p_i}$.
- β_0 refers to the intercept term, the average log-odds of the specific result.
- $\beta_{\text{Position}_{\text{positionid}[i]}}$ refers to the position player effect coefficients for the relevant position player $[\text{positionid}[i]]$ in possession i .
- $I_{\text{positionid}[i]}$ is an indicator variable equal to 1 if the relevant position player $[\text{positionid}[i]]$ touched the ball in the possession, and 0 if not.
- $\alpha_{\text{Team}_{\text{teamid}[i]}}$ refers to the team effect for the relevant team $[\text{teamid}[i]]$ in possession i .

Hierarchical models account for grouping structured by allowing parameters to vary by subgroup. In this case, random effects are used for each position with normal distribution. Player effects are drawn from their respective positional distribution.

The results give the mean for each position and player. These values do not give the importance of each position, but rather how win probability changes per touch. For example, middle blockers have a very high coefficient - but this reflects a strategy where middle blockers are set only after an excellent reception. A touch by the middle blocker is one made under optimal circumstances.

As an extension of this, Fellingham introduces the metrics Points Above Average Per Set (PAAPS) and Points Above Average Per 100 touches (PP100). By substituting a player into an 'average team', they note the change in point win percentage. This is scaled by the number of touches per set or per 100 touches respectively. This is conducted both for the increase in positive outcomes and reduction in negative outcomes.

Insights

The results have moderate accuracy in assessing award winners. However, there are occasional disparities where some award winners are viewed as a significantly negative influence. Furthermore, outside hitters typically have very low impact, which is antithetical to coach sentiment as one of the most influential. The best players according to PAAPS typically have $\sim 1 - 2$ points of impact per set.

Future extensions could account for defensive impact or opponent presence. Whilst the

technique does model both positive and negative outcomes, defence is largely based around influencing the opponent. A skilled blocker in the right position should prevent highly aggressive spikes from being attempted. Defensive back-court players may generate additional digs that would have otherwise scored a point. Modelling 'non-events' and changes in the opponent's chain may boost the ratings of defensive players such as the libero who appear least impactful in the model.

3.4.2 Points Gained Per Set - Markov Chain Model

Baseball features a popular methodology referred to as RE24. A value for each state in baseball is computed. Player impact is then assessed based upon how they change these states. Given the discrete touches in volleyball, a similar methodology could be applied. Powers *et al.* [40] uses this change in point-win percentage after each action as an indicator of player ability.

A Markov approach is taken to produce the point-win probability (PWP) of each state. States are defined by the prior action and its grade. A k-step transition is used to find the likelihood of moving from some state s to s' in k-steps. In this case, a 100-step transition based on empirical data defines the probability the receiving team eventually wins the point. Thus, each action has an associated PWP.

As an example, a poor pass (R-) may have a PWP of 40%. A setter may convert this to a good set (S+), which has a PWP of 70%. Thus the setter increased the PWP by 30%, worth 0.3 points.

The other key concept - and where the bulk of the paper lies - is the strength of schedule (SOS) model. Actions taken to score against the other team are treated as Adversarial Contact Groups (ACGs). This includes the serve/reception group and the attack/block/dig group. Generalised linear models and hierarchical random effects for players, teams, and conferences are used to determine these outcomes. In this way, teams are regressed towards conference and players towards teams.

As an example, change in win probability for the serve is modelled in (3.5). The $v(s)$ values refer to the PWP of the serving team winning the point.

$$y_i = -(v(s'_i) - v(s_i)) \quad (3.5)$$

The y_i value is treated as a realisation of the random variable Y_i , modelled with a normal distribution incorporating hierarchical effects. The expression for this distribution is given in (3.6). Conference, team, and the player effect for serving are given by γ_{c_i} , τ_{t_i} , and π_{p_i} . Similarly, $\tilde{\gamma}_{\tilde{c}_i}$, $\tilde{\tau}_{\tilde{t}_i}$, and ρ_{r_i} refer to the random effects of the opposition's conference, team, and player effect for receiving.

$$Y_i \sim \text{Normal}(\eta_i, \sigma_\varepsilon^2) \quad (3.6)$$

$$\eta_i = \alpha + \gamma_{c_i} + \tau_{t_i} + \pi_{p_i} + \tilde{\gamma}_{\tilde{c}_i} + \tilde{\tau}_{\tilde{t}_i} + \rho_{r_i}$$

Hence, the estimated strength of server p_i is given by $\tilde{\gamma}_{\tilde{c}_i} + \tilde{\tau}_{\tilde{t}_i} + \tilde{\pi}_{p_i}$. This is the SOS for receiver r_i . In summary, the transcript of all contacts and their change in PWP are used to estimate the underlying SOS distribution of the agents involved in the ACG.

A similar process is conducted for the attack/block/dig chain. However, it is far more complex and will be summarised. A chain of attack states and their outcomes is produced. A unique normal regression is estimated for each element of the chain. For example, an attack error is not assumed to be influenced by the defence, and does not include defenders. Blocker and digger responsibilities are inferred separately based on attack zone.

The previous two methods are combined to generate the final points gained metric. For most actions, this process simply credits the individuals with the change in PWP based on the SOS. For the attack/block/dig however, the contributions of the attacker and setter are split. The variance explained by the random effects for each party is used to divide their contribution.

Insights

This study has multiple findings. The first is the distribution of each action on the change in PWP. These take roughly normal distributions, with the best performers in these categories having about $\pm 5\%$ PWP in these skills. The exception is setting, which is viewed as largely neutral. The model learned that attacker identity explains more variations in attack outcome than setter identity.

Notably, this approach is used to derive a points gained per set metric. The highest in this category are outsider hitters, as would be expected due to their high number of touches. Elite players are contributing an advantage of roughly 1 point per set. The matchup between award winners and points gained per set is highly aligned.

An element to explore is the setter is found to be almost entirely neutral, contrary to domain knowledge. The setter should make a huge impact on team performance, as hitting ability is highly influenced by setting ability. It may be that setters at this level have reached a degree of consistency where it is simply up to the hitters to convert the attack.

3.4.3 RAPM - Bayesian Model

RAPM has promising application in volleyball. Awosoga implements such a technique [41]. As contributions to the field, a Bayesian methodology is used, as well as incorporating the contextual factor of score closeness.

A horseshoe prior is used for the Bayesian methodology. This is suited for the sparseness of the player matrices used in RAPM calculations. An additional input of score closeness is added to account for the tendency for blowout games to be played less aggressively.

Insights

A biasing is noted towards the libero in terms of PM. This is partially due to substitution tendencies, where liberos are off the court for 2 out of 6 serves. It has been shown the serving team generally loses the point - which the libero will not be present for.

Comparing with award winners, Awosoga notes these players are highly positive in RAPM scores with values ~ 0.18 . Converting from log odds, this is $\sim 4.5\%$ chance increase in winning each point. Across 50 possessions, this is approximately a 2 points per set contribution.

While Awosoga's model is conceptually similar to existing RAPM frameworks, it provides a useful benchmark for comparison. Elite players seem to contribute ~ 1 to 2 points per set. Any outlier impact for liberos should dually be noted.

3.4.4 Credit above Value Expected - Markov Chain Model

Hass completes a PhD dissertation on the topic of division of credit modelling [42]. A novel framework is described. The technique is intended to be sport-agnostic and generalisable to any sport. Furthermore, an extensive literature review is conducted on the topic.

As a preliminary to the model, Hass implements an RAPM model for a women's volleyball team. A significant part of this process involves the recovery of court information - primarily for the libero. He and Craig have written a companion paper on the topic [43]. These methods demonstrate a viable framework for accurate court recovery. In their implementation of RAPM, ridge and lasso regression are used. Additional logistic methods such as Bayesian logistic regression and regularised logistic regression are included. These hold similar results, but are strong options for PM models given the point structure in volleyball.

In preparation of their model, they describe the inputs to the system. Actions are graded independently of result and have 3 grades - advantageous, neutral, and negative. For example, a poor serve will be rated poorly even if it scores a point due to an error in reception. Hass then builds a transition matrix using Markov chain modelling. The eventual matrix gives the probability of a team eventually winning the point after each action. A Bayesian approach with Dirichlet prior is used to calculate these probabilities.

The latter half of the paper takes a very unique approach to impact evaluation. Each action contributes an amount equal to its Markov derived chance of eventually winning the point. For example, a player passing neutrally (+0.5), then subsequently hitting the ball advantageously (+0.7) will receive +1.2 'credits'. Once the point is realised, the credits earned by each member of the team are totalled. Players' relative credit is thus a representation of how responsible they are for each point. A player responsible for 1.2 credits out of 5 is responsible for 24% of the point outcome. Court presence is a constant factor, netting a $\frac{1}{42}$ credit value for every touch. Contextual elements - such as service, homecourt, and other miscellaneous factors - are also included in the model.

To evaluate relative performance a figure for the baseline contributions of each position is necessary. Hass builds a Value Expected Model, which first generates the expected score and subsequently how often each role contributes. Logistic regression is used to estimate the win probability for a point. Roles are then calculated using their distribution of actions. For example, players setting frequency are likely playing setter. The role allocation portion is an extensive part of the paper to account for unique strategies and player archetypes.

Player evaluation is identified in the ultimate model: Credit above Value Expected (CaVE). This is calculated as in (3.7), the difference between a player's observed credit and their expected value. A comparison with APM yielded significantly different results.

$$\text{CaVE}_k = \text{Credit}_k - \text{ValueExpected}_k \quad (3.7)$$

Simulation was used to validate the CaVE model. Interestingly, this was through the use of a video game, which features numeric values for each player skill. The evaluation scores can be compared against the actual latent skill designated in the program. The average of a player's numbers was taken to determine overall player ability. In terms of player ability, a random guess was superior 18.2% of the time. However, for exact skills, a random guess was superior only 1.49% of the time.

Insights

This dissertation presents a highly original technique for player evaluation. The methodology for court presence recovery is particularly valuable for future modelling. Using simulation to verify models is an irregular - but perhaps useful - approach to model verification. Furthermore, a meshing of both player presence and player action is evident. However, there are some concerns with point loss allocation and the derivation of expected value. The use of touches as an indicator may be misleading. For example, a hitter may make a strong attack whilst eventually losing the point. This appears worse than making no hit at all despite the effectiveness of the attack.

3.4.5 Summary of Volleyball Player Impact Evaluations

Through this study, some clear benchmarks of player effectiveness present themselves. Most of these metrics agree elite players contribute ~ 1 -2 points per set. Whilst presence-based models (such as RAPM) have a relatively even distribution of player impact, those based upon player actions tend to place a small value on the libero. This may stem from an inability to model 'generated events' such as from good positioning. The topics, approaches, and useful takeaways from these papers are summarised in Table 3.2.

Table 3.2: Analytical Papers Specifically Evaluating Volleyball Player Impact

Author	Statistic	Findings	Evaluation
Fellingham [1]	Bayesian hierarchical logistic	Elite players contribute ~ 2 points per set. Liberos make the smallest impact on possession outcome.	An effective translation of PM techniques to volleyball. Lacks defensive statistics incorporating the other team.
Powers <i>et al.</i> [40]	Markov chain modelling and hierarchical linear models for action outcomes	Elite players affect win percentage by $\sim 5\%$ per touch, summing to ~ 1 point per set. Setters play the biggest role in whether there is a clean attack or block touch.	Highly tailored, interpretable model including offensive and defensive skills. However, relies heavily on graded actions and views the setter as neutral.
Awosoga [41]	RAPM with Bayesian methods and horseshoe prior	Elite players contribute a $\sim 4.5\%$ increase in win percentage with each touch. Liberos may have outlier presence-impact statistics due to the serving team's propensity to losing which they are not present for.	A well-structured implementation of RAPM concepts into volleyball.
Hass [42]	Markov chain modelling and custom division of credit method	In-depth and accurate court presence reconstruction	Interesting approach that has not been applied to elite players and displayed some correlation with player skill.

3.5 Towards a New Model

In the wake of the literature review we can form some conclusions. First are some benchmarks of maximum player impact. The best players generally reside in the vicinity of ~ 1 - 2 points. Moreover, elite performers of each skill contribute a roughly 5% increase in rally win probability. Markov and Bayesian regression methods have seen effective and insightful application.

The reviewed literature highlights several promising directions. Of particular note is Statistical Plus-Minus. Using RAPM as a benchmark of player impact, we may use various player statistics to predict this value. Not only would this produce the more stable SPM value, it is possible to synthesise new statistics for player performance. One such statistic is 'spike strength' - a hitting metric not only incorporating direct point-scoring outcomes, but the opponents ability to return the ball. Another to measure reception ability may be the team's hitting efficiency after a particular player receives the ball.

However, the approach proposed is an extension upon Fellingham's work with touch-based modelling. Whilst their proposed methodology demonstrated effectiveness and broached new ground, it is fairly general. By incorporating the specific skills of volleyball - and a host of other modifications - into the model, improved accuracy and data granularity is pursued in this study. Furthermore, we can compare the accuracy and legitimacy of the technique to more traditional RAPM values.

Chapter 4

Regression Methodology

4.1 Data

The dataset for this project has been procured from a single season of a publicly available source that will remain anonymous. For the purposes of anonymity, player teams and names have been replaced with a random set. In this case, teams are referred to by a single letter (E.g. "A"), and players on that team by a random name beginning with said letter (E.g. "Anthony", "Aaron", etc).

A web scraper was developed in PyCharm using the `BeautifulSoup` and `playwright` libraries. This extracts the play-by-play information from this league. No .json files could be tracked, and so the information was extracted from the html directly. This code is provided in Appendix B. Whilst the initial rotation lineups and in-game substitutions were provided - the point-by-point rotation was not displayed. As such, the rotation was reconstructed based on point outcomes. This is a fallible method, and several games were detected as having illegal rotation states. I.e. Libero in the front-court. The dataset was reduced to exclude these games, and the set further reduced to games involving a select 10 teams.

Player role was extracted based on listed player position - also retrieved with scraping techniques.

Each match has been collated into a .csv file of the touch-by-touch sequence of the game. A row corresponds to a single touch including the following: Player identity, skill being performed, direct point-winning outcome, player team, current possession, and various game context elements such as the serving team and current player rotation.

The major information categories for the present work are the following:

- Player, team, and role. This is for identification.
- Skill and point-winning outcome. This labels the skill and any direct points or errors.

- **Possession.** This is an indicator for consecutive touches by the same team.

All data processing after data extraction was conducted in RStudio. The data was modified with an additional facet referred to as 'possession outcome'. It indicates the direct point-winning outcome for that possession. Possession outcome is recorded as a +1 for a direct point, a -1 for a direct error, and a 0 for play continuation. A direct point is one where the next action is an error. Take the following situation: A point is scored off a block or dig error. The possession outcome for the attacking team is +1, and the possession outcome for the defending team is -1. However, a dig followed by a setter error is strictly recorded as a -1 for the team in possession.

The other major information categories for the present work includes the lineup configuration, score, and serving team. The lineup configuration indicates the rotation position each player must be in during service, as well as the setter's current position.

To illustrate the key data used, an example rally is provided in Table 4.1.

Table 4.1: An example sequence of touches from the dataset

Player	Team	Role	Skill	Outcome	Possession	Poss. Outcome
Aaron	A	Outside	Serve		51	0
Bobby	B	Libero	Reception		52	0
Bella	B	Setter	Set		52	0
Bismack	B	Opposite	Attack		52	0
Anthony	B	Outside	Block		53	1
Astarion	A	Libero	Dig		53	1
Angus	A	Setter	Set		53	1
Alex	A	Middle	Attack	Point	53	1
Bobby	B	Libero	Dig	Error	54	-1

This dataset covers 17,952 points and 233 players.

For the possession outcomes, there are 52,124 entries, of which 13,133 hold +1, 24,371 hold 0, and 14,620 hold -1.

4.2 Analytical Approach

This study conducts exploratory research into the factors influencing regression-based models for division of credit in volleyball. Three primary regression models were used as a basis to evaluate the players in the data. From this basis, various modifications were made to evaluate potential benefits and properties of these techniques.

- **Regularised Adjusted Plus-Minus (RAPM):** This first is a presence-based approach using linear regression with regularisation. This is an implementation of RAPM, which boasts considerable literature.
- **Touch Plus-Minus (TPM):** The second is a touch-based approach using logistic regression with regularisation. This mirrors the work of Fellingham [1].
- **Skill-Touch Plus-Minus (STPM):** The third extends upon this using a skill-touch-based approach, similarly using logistic regression with regularisation. Several opportunities for analysis open up by denoting the skill.

The presence-based approach features 4 different modifications, the touch-based approach 2, and the skill-touch-based approach 6 variations. These modifications are performed both independently and in combination.

This is an exploratory process to identify which factors are most important for most accurately determining match outcomes. By comparing how these augmentations affect the models, insight can be gained into which aspects most explain a volleyball game, and how individuals can impact the outcome. This analysis aims not only to identify the most accurate model, but also to provide insights into the statistical properties of volleyball.

Models will be analysed in 4 main facets:

- **Comparison with Team Success:** Evaluates the extent to which a model can infer overall team performance from individual player estimates.
- **Prediction Accuracy:** Assesses the precision of a model's predictions based on cross-validation.
- **Model Biasing:** The model's learned values for various positions, skills, and contexts.
- **Data Granularity:** A model's ability to decompose a player's impact into specific contexts.

Data granularity is not measured specifically, but rather a qualitative value. In analytics, it is useful to not only know the extent of a players impact, but the situations where this impact is made. For example, between the front-court and back-court for substitution.

4.2.1 Regression Technique

Prior applications of regression in volleyball have used Bayesian techniques. These are powerful as they not only provide a posterior distribution, but effective and flexible shrinkage parameters. Due to the large selection of models that will be constructed and run, a frequentist methodology will be used. Linear and logistic regression with regularisation will be used. Given the number of models considered, speed and simplicity is preferred over the slightly more intelligent shrinkage of Bayesian methods. These computations will be performed courtesy of the `glmnet` package.

4.2.2 Data Import

From the collected .csv, it is relatively simple to merge each game. The dataframe is produced.

```
aggregate_files <- map(game_files, extract_file)
aggregate_plays <- map_dfr(aggregate_files, process_positions)
```

This operation simply reads the .csv files into R, whereupon they are combined with some additional processing. E.g. Normalising naming scheme, applying listed player positions,

4.3 Regression Structure

Regression based modelling relates player participation to point-outcomes. To perform this, we need both a design matrix and outcome vector. Each row of the design matrix represents the players involved in the observation. Each entry of the outcome vector represents the outcome of this observation. Building these design matrices directly from the data is difficult - and so it is transformed into an object which will be referred to as a 'segment' object. This is a reduced dataframe which is easier to convert directly into a matrix object.

4.3.1 Observation Segments

The objective is to transform the touch-by-touch information into observation-by-observation information. One row should correspond to one observation.

A simplified implementation for the point-by-point segments is provided in the following section of code:

```
point_segments <- aggregate_plays %>%
  filter(point == TRUE) %>% # Reduce observations to a per rally basis.
  # Produce the outcome of each observation.
  mutate(value = ifelse(point_won_by == home_team, 1, -1))
```

The resulting dataframe lists the players on court for each point and the outcome. An example excerpt is given in Table 4.2. For touch-based segments, the process is moderately more complex. However, the result is similar in a dataframe listing the players who touched the ball and the outcome.

Table 4.2: An example excerpt from the presence segments

H_1	...	H_6	V_1	...	V_6	Value
Aaron	...	Alex	Bobby	...	Bella	1
Aaron	...	Alex	Bobby	...	Bismack	-1
Astarion	...	Alex	Bobby	...	Bismack	1
Astarion	...	Alex	Bobby	...	Bismack	-1

Note: Slot H_1 refers to home player 1; V_1 to visiting player 1.

Similarly, an example sequence from the touch segments is given in Table 4.3.

Table 4.3: An example excerpt from the skill-touch segments.

Par1	Par2	Par3	Par4	Value
Aaron				0
Bobby	Bella	Bismack		0
Anthony	Astarion	Angus	Alex	1
Bobby				-1

Note: Slot Par1 refers to participant 1.

4.3.2 Player Tagging

A heavily used technique in this project was the application of tags to differentiate contexts from players. Each player-tag combination is treated as a separate entity. Note that this increases the sparsity of data when performed, as the number of predictors will multiply for each tag.

As an example of this, player touches can be converted to 'skill-touches', which include the skill the player was performing. The application is very simple in code:

```
df %>%
  mutate(skill_player = paste(skill, player_id, sep = "|"))
```

The resulting set of segments is given in Table 4.4. Each row corresponds to a possession, listing the participants (Par), and the possession outcome.

Table 4.4: An example excerpt from the skill-touch segments

Par1	Par2	Par3	Par4	Value
Aaron Serve				0
Bobby Reception	Bella Set	Bismack Attack		0
Anthony Block	Astarion Dig	Angus Set	Alex Attack	1
Bobby Dig				-1

Note: Bobby|Reception indicates Bobby performed a receive. This is entirely distinct from Bobby|Dig in the regression.

4.3.3 Matrix Construction

The next step is to construct the matrices. The following code segment produces the player matrix for point-based segments:

```
p_trix_pm_raw <- produce_player_matrix(point_based_segments, "H_", "V_")
```

It produces a sparse matrix with entries for involved players. We can observe the non-zero entries for observation 6.

```
observation6 <- p_trix_pm_raw$X[6, , drop = FALSE]
nonzero_cols <- which(observation6[1, ] != 0)
named_values <- observation6[1, nonzero_cols]
names(named_values) <- colnames(p_trix_pm_raw$X)[nonzero_cols]
named_values
```

Villegas	Jordan	James	Vixen	Jerry
1	-1	-1	1	1
Victor	Valentine	John	Jack	Virgil
-1	-1	1	1	-1
Valery	Jackson			
-1	1			

Players with an entry 1 are home players, and those with -1 are visiting players.

This process is repeated for each observation set. Both the design matrix and observation vector are attached to this `p_trix_pm_raw`, as well as additional information such as the intercept, penalisation term λ , and the cross validation model.

4.3.4 Regression Computation

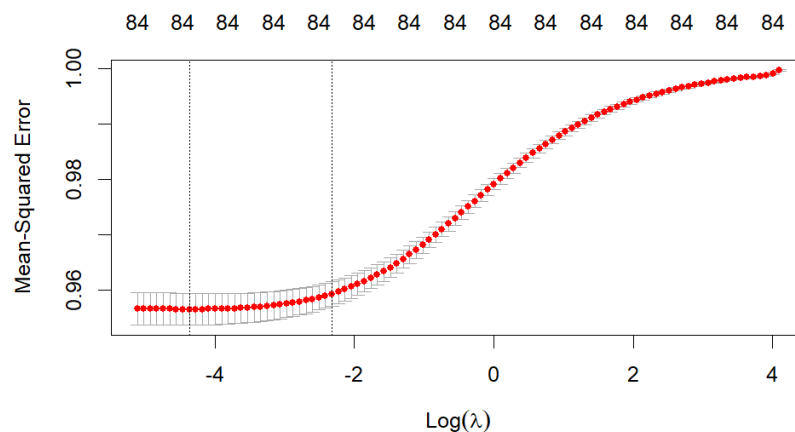
Presence-Based Models

A linear ridge-regression technique following an RAPM style approach was used. The techniques implement the methodology from equation (3.1) and minimising cost function (3.2).

Choosing the appropriate regularisation parameter λ is an important factor in model accuracy. In this case, it was performed through cross validation in the `glmnet` package.

The following code segment plots the mean-square error for out of sample prediction based on λ .

```
cv_model <- cv.glmnet(p_trix_pm_raw$X, p_trix_pm_raw$Y, alpha = 0)
plot(cv_model)
```



The optimal value of λ can then be extracted simply using the minimum.

```
best_lambda <- cv_prep$lambda.min
model <- glmnet(p_trix_pm_raw$X, p_trix_pm_raw$Y, alpha = 0)
```

This produces an RAPM value for each player in the design matrix. The top 5 performers will be displayed, filtering for those with more than 1000 possessions.

	player_id	RAPM_raw	points_played
1	Gary	0.409	2230
2	Paul	0.386	2099
3	Benjamin	0.363	2471
4	Jesse	0.336	3071
5	Alucard	0.316	2744

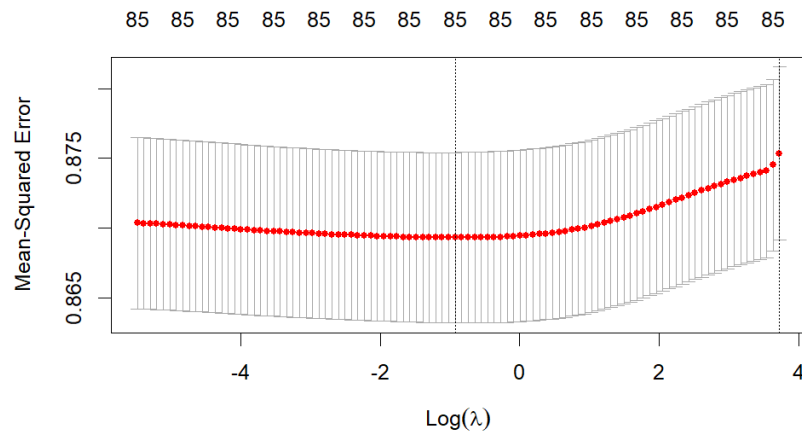
These numbers are incredibly large. Each outcome is split between +1 and -1, so to convert to change in point-winning percentage the value must be multiplied by 50. This indicates top performers influence point-winning outcomes by $\sim 20\%$, a colossal value. This is not unexpected however, and most papers find liberos to skew results in this fashion due to their unbalanced serving participation [41]. The libero is generally on for 4 serving possession and 6 receiving possessions. In this dataset (and most professional leagues), the receiving team wins $\sim 70\%$ of the time.

4.3.5 Illustrative Example

To demonstrate the modelling workflow, the ridge-regression process will be repeated - with an augmentation. Clearly, some serving discrepancies are being captured. The design matrix will be modified to include the additional column `serving_adj`. This simply holds +1 if the home team serves, and -1 if the visiting team serves.

When re-running the cross-validation, the results are much more stable. The serving adjustment is also removed from the penalisation term.

```
plot(cv_model_serving_adjusted)
```



The player scores are similarly far more reasonable, at around 3% per player per point for these best performers.

	player_id	RAPM_raw_sadj	points_played
1	Jerry	0.0381	2641
2	Zion	0.0337	1795
3	Perry	0.0279	2407
4	Zagreus	0.0245	2208
5	Alucard	0.0231	2744

This process also gives us the effect of serving on point-outcomes.

serving_adj	-0.398	19671
-------------	--------	-------

As expected, serving is incredibly disadvantageous, reducing the chance of winning the point by $\sim 20\%$. It is hypothesised that liberos were absorbing a significant amount of the serving effect. Intuitively, if the libero is not on the court (team is serving), the team only has a $\sim 30\%$ chance of winning the point. The model sees this, and gauges that the libero is an incredibly important player.

Touch-Based Models

Logistic models are limited to binary outcomes, yet three possession outcomes are possible (point, continuation, error). Hence two regressions are run: One using only positive outcomes (negative outcomes treated as 0), and the other using only negative outcomes (positive outcomes treated as 0). An effective player should improve the log-odds of a positive outcome and reduce the log-odds of a negative outcome. The following model (4.1) is used.

$$P(Y = \text{result} \mid X) = \frac{1}{1 + e^{-\hat{Y}}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}} \quad (4.1)$$

Where the *result* term takes the value +1 for positive outcomes, and -1 for negative outcomes. This functions identically to the logistic regression equation given in Equation (2.3). A 1 in the player matrix X_{ij} indicates that player i touched the ball in possession j . All other players are a 0 for that possession.

Furthermore, ridge regularisation techniques are applied - applying the cost function expression as in Equation (3.2). Using the standard implementation, the penalty is applied for difference from 0. It is possible to adjust this by position to alter the penalisation setpoint. Reasonable values for this could be sourced from Fellingham's paper, as they found the posterior averages for each position in their dataset. This however has not been performed, and a blanket regularisation of difference from 0 has been implemented.

The following code segment produces each player's log-odds effect on possession outcomes. A modified Y term uses either positive outcomes or negative outcomes.

```
cv_model <- cv.glmnet(p_trix_pm_touch_raw$X, p_trix_pm_touch_raw$Y,
                     alpha = 0, family = "binomial")
best_lambda <- cv_prep$lambda.min
model <- glmnet(p_trix_pm_touch_raw$X, p_trix_pm_touch_raw$Y,
               alpha = 0, family = "binomial")
```

The top 5 performers in terms of improving positive outcomes will be found, filtering for those with more than 50 touches:

	player_id	TPM_raw_pos	TPM_raw_neg	total_touches
1	Antonio	1.33	0.828	2379
2	Patrick	1.32	0.920	906
3	Jackson	1.31	0.404	785
4	Penelope	1.22	0.718	882
5	Bella	1.19	0.907	1599

These represent a player's influence on the log-odds of scoring a point. These top performers have an influence ~ 1.4 . Also note the sizeable influence on negative outcomes of these players. A baseline is required to assess the percentage effect on possession-outcomes. Assuming a baseline 50% (log-odds 0) chance of scoring a point, removing these players from the observation will drop the log-odds by -1.4, resulting in a mere 20% chance of scoring a point. Whilst this may seem extreme, each of these players is a setter. If the setter does not touch the ball, it may indicate an error has occurred in the play. To explore this, the intercept for the regression is found as well as the average value for positive outcomes.

```
attr(TPM_raw, "pos_attributes")$intercept
```

```
[1] -3.092571
```

```
> TPM_raw %>%
  summarise(avg_pos = mean(TPM_raw_pos, na.rm = TRUE))

avg_pos
0.792
```

Clearly, the baseline for scoring a point is very low, and multiple touches are needed to win a point.

Furthermore, we can compare our results to those conducted by previous attempts and compute the weighted mean for each position. These values, and those found by Fellingham [1], are provided in Table 4.5. Clearly there is a strong resemblance for the positive case, and the general trends remain the same. The intercept is highly negative, and the ranking of positions is roughly similar. I.e. Setter is highest, followed by the middle.

Note however that these are not directly comparable. The datasets are different, and shrinkage has been applied differently - where Fellingham uses hierarchical modelling as opposed to a blanket regularisation in this paper.

Additionally, the criteria for positive and negative outcomes has been defined slightly differently. The methodology of this paper uses a 'direct' point outcome - occurring in the subsequent action. Fellingham however uses the entirety of the next play. This may explain the discrepancies between the averages found for negative outcomes.

Table 4.5: Comparison of role-wise average coefficients between this study and that of Fellingham [1]

Role	TPM⁺	Fellingham⁺	TPM⁻	Fellingham⁻
intercept	-3.0925	-2.801	-0.204	0.254
libero	0.866	0.572	-0.0470	-0.094
middle	0.969	0.770	-0.429	-1.043
opposite	0.869	0.651	-0.293	-0.722
outside	0.837	0.671	-0.323	-0.809
setter	1.219	1.329	-0.751	-1.248

Note: TPM⁺ refers to the coefficient for positive outcomes; similar for TPM⁻

Chapter 5

Model Augmentations

The bulk of this project investigates how augmentations may influence the results to improve accuracy. The modifications and their brief descriptions are given in Table 5.1.

Table 5.1: Overview of model types, their augmentations, and a brief description.

Model Type	Augmentation	Description
Presence-Based	BASE	Regresses court-presence against point outcomes
	FAB	Front-court and back-court status
	sadj	Serving adjustment
	HitNet	Shifts observations to hitting-outcomes
	OD	Shifts observations to hitting-outcomes, separating offensive and defensive possessions
Touch-Based	BASE	Regresses touch-presence against possession outcomes
	FAB	Front-court and back-court status
	OPP	Includes opponent court-presence
Skill-Touch-Based	BASE	Regresses skill-touch-presence against possession outcomes
	FAB	Front-court and back-court status
	OPP	Includes opponent court-presence
	stone	Sum-to-one. Normalising design matrices to total to one.
	prevact	Includes the previous opponent skill-touch
	prevseq	Includes the previous opponent possession (sequence of skill-touches)
	phitters	Presence-hitters. Uses court-presence of available hitters rather than Attack skill-touches

5.1 Touch-Model Conundrums

Upon consideration, it is hypothesised that a touch-based model will suffer from several issues. The overlying issue is that successful and unsuccessful actions will modify the structure of the player matrix itself - rather than just the outcome. In other words, the player matrix is endogenous. These issues will broadly be categorised as error conditions and participation bias.

5.1.1 Error Conditions

Errors will typically end the play immediately. The case-study for this issue will be the dig. Recorded in the dataset are 9375 digs, of which 3041 occurrences feature the dig as the only touch in the possession. Of these 3041 single-observation digs, 78.8% are errors (the remaining are sent back over the net). This is an expected relationship, as a successful dig will keep the ball in play and an error will not. From a regression standpoint a single-touch possession presents a misleadingly definitive case; one action and one outcome. As a result the model learns the inherent value of a dig is highly negative. Conversely, when a dig is successful, subsequent actions (such as sets and attacks) are observed and associated with positive outcomes. The regression will thus attribute extreme positive value to the ensuing actions, as theoretically they transform the 'awful' dig into a positive play.

Consequently, a players genuine ability to dig the ball may not be captured by the model at all. As the failure conditions will teach the regression their 'true' value, and successes are entirely attributed to the other actions 'rescuing' the play.

Generating Opportunities: Additionally, a players ability to 'generate' touches can manifest itself here as well. Consider a lazy player, who only digs balls that are easy. Out of 300 potential digs, they successfully complete 30 with no recorded errors. In contrast, an excellent, active libero may successfully dig 80 of 300, with 40 errors. The model will consider the lazy player to have a much more effective dig, despite the libero generating many more opportunities for their team.

5.1.2 Participation Bias

Not only are the number of agents dynamic, but the types of players. The case-study for this will be the middle and their attacks. Generally, middles are only set after a good pass. As a result, they are only set under ideal circumstances. This pattern reflects empirically. We find the number of attacks and hitting percentage for each role.

	role	total_attacks	kills	errors	hitting_percentage
1	middle	3400	2025	329	0.499
2	opposite	5966	2944	963	0.332
3	outside	9843	4678	1481	0.325
4	setter	245	121	22	0.404

As shown, middles are set far less often, but have a much higher hitting percentage. On the other hand, outsides are the 'outlet' option for if the system falls out of play. A result much like the 'error condition' can manifest.

An extreme example will be provided. Assume all good passes are set to the middle, and all other passes set to the outside. Now we simulate example player Jessica. From an objective standpoint, she may be fantastic in reception. After she passes, the team has a hitting percentage of 47%. However, from the models perspective - the most important player to look at is the attacker. When the middle is present, the team wins 49.9% of the time, and for the outside, the team wins 32.5% of the time. From the model's perspective, Jessica's presence has zero impact. Her contribution will likely be measured as how much the individual hitters improve. For example, after she receives, the players hit 2.5% better (52.4% for the middle, and 35% for the outside).

The ideal measurement identifies an adjusted measure of how the team performs when she passes. In this dataset, the hitting percentage is 38.6% for an attack after reception. For Jessica, her value of 47% is phenomenal. But it may not be attributed to her, but rather to her hitters.

Forced Actions: An extremely similar case is when actions are taken out of necessity. For example, a poor reception may necessitate a non-setter to execute the set. Hence, most non-setters only set the ball in unfavourable situations and subsequently receive a negative evaluation for setting actions. These players should not be evaluated poorly for these forced actions.

5.2 Presence-Method Conundrums

Presence-based methods of player evaluation are popular and effective. Volleyball has high pace and relatively frequent substitutions which are highly suited for this type of analysis. However, the substitution patterns in the sport may lead to structural inaccuracies in the model. Firstly, players are almost always substituted by role (E.g. Setters are substituted for other setters). Secondly, it is common for players to play more than 95% of points.

Perfect Collinearity and Backup-Relative Rating

These two factors may lead to very poor on-off signalling. It is entirely plausible for players in the same role to share 0 court-time, and perhaps most pertinently, where at least one of these players is on court at all times. This is a form of perfect collinearity. Without observations where both players are absent, their independent contributions cannot be disentangled statistically. Furthermore, RAPM metrics tend to become noisy when subject to a small 'off' signal.

Structurally, the score a player receives may be relative to other players of that role on the team. Take an example: A team has two strong middles of equal skill, and team succeeds

because of them. However, since the team performs equally well independent of which middle is in, their impact may appear to be 0. Considering this applies for every players and positions, a player’s RAPM score may describe an impact relative to their teammates sharing the same role, rather than an absolute impact.

Team Construction Example

For this dataset, of Team B’s 4676 points played, 4586 (98.1%) were played by their main setter. The extremely small ‘off’ sample is the only criterion to evaluate their impact separate from the team’s success. And furthermore, this can only be taken relative to the other setter who played in the remaining rallies.

Despite this, these RAPM values will be treated as an acceptable baseline of player quality. Holistically, they still represent how the team performs with said player.

5.3 Presence-Based Model Augmentations

5.3.1 Point-Based RAPM

A point-based RAPM approach has seen several implementations in volleyball [41] [43]. This typical method has been emulated using simple regression. This is the basic model that has been implemented and will be augmented.

5.3.2 Hitting-Based Observations (HitNet)

Motivation

Hitting efficiency serves as a strong analogue to basketball’s offensive and defensive rating. Furthermore, it is a highly predictive indicator in volleyball [28] and should function as a suitable measure of match outcome. Rallies can include multiple hitting observations, potentially giving an expanded dataset to draw from. From the dataset, $\sim 74\%$ of rallies are decided by an attack, $\sim 23\%$ decided by the serve, and the remaining 3% from various errors. E.g. Setting, freeball.

This modification serves to assess the baseline accuracy of shifting to hitting based observations.

Implementation

The augmentations are relatively simple. Rather than using point outcomes as observations, hitting outcomes are used. Three outcomes are possible: Point (+1), continuation (0), and error (-1). Linear regression is still used over logistic, as hitting percentages are generally stable.

Expectations

This score will largely mirror a purely presence-based approach, and may exhibit slightly better stability.

5.3.3 Hitting-Based Observations with Offense and Defense Split (OD)

Motivation

An advantage of using hitting efficiency is separating a player's influence on offensive and defensive possessions. This may highlight where a player's strengths lie. However, given the extremely dependent relationship between defense and offense, this may be a faux interpretation. A strong offensive player should reduce the opponent's ability to counterattack, thereby improving the team's defense. Hence, it can be compared with the previously discussed net influence on hitting to note any insights. Furthermore, we can explore the correlation between a player's offensive and defensive rating.

Implementation

For each hitting segment, the offensive and defensive players are given the **OFF** and **DEF** prefixes respectively as in Table 5.2.

Table 5.2: An example excerpt from hitting segments where offense and defense have been differentiated

H_1	H_2	...	V_5	V_6	Value
OFF Aaron	OFF Alex	...	DEF Bobby	DEF Bella	1
DEF Aaron	DEF Alex	...	OFF Bobby	OFF Bismack	1
OFF Astarion	OFF Alex	...	DEF Bobby	DEF Bismack	0
OFF Astarion	OFF Alex	...	DEF Bobby	DEF Bismack	-1

Expectations

This score will largely mirror a purely presence-based approach. The offensive and defensive scores will show some degree of positive correlation.

5.3.4 Front-Court and Back-Court (FAB)

Motivation

Highly unique to volleyball is the concept of rotation. The rotation changes regularly, and the responsibilities and rules for players changes distinctly between the front and back-court. The disparity of a player between front-court and back-court performance is perhaps the most important knowledge for a coach in terms of substitution.

Implementation

Implementing this split was done by differentiating between the front-court (positions 2, 3, and 4) and back-court (positions 1, 5, and 6) versions of a player. The updated segments are given in Table 5.3.

Table 5.3: An example excerpt from the front-back separated presence segments

H_1	...	H_6	V_1	...	V_6	Value
FT Aaron	...	FT Alex	FT Bobby	...	FT Bella	1
BK Aaron	...	FT Alex	FT Bobby	...	FT Bismack	-1
BK Astarion	...	FT Alex	BK Bobby	...	FT Bismack	1
BK Astarion	...	BK Alex	BK Bobby	...	FT Bismack	-1

Expectations

Expected to yield significant performance improvements, as rotation lineups significantly affect rally outcomes [29].

5.3.5 Serving Adjustment (sadj)

The serving adjustment has been previously discussed. The `serving_adj` term is unpenalised, and so the cost function for the model is given as in 5.1. Serving should not 'share credit' as players do.

$$\sum_i \left(Y_i - \beta_0 + \beta_{serve} \times I_{serve,ind} - \sum_{j=1}^n \beta_j X_{ij} \right)^2 + \lambda \sum_{j=0}^n (\beta_j)^2 \quad (5.1)$$

Expectations

Model accuracy and predictive power will improve in all cases.

Limitations

Most modern RAPM techniques adjust for home-court advantage in some capacity. Whilst serve-receive is analogous to home-away (i.e. Measurable advantage to one team), there is a clear structural difference between the serving and receiving team. Consequently, some teams may exhibit clear strength in this area. It may be more accurate to allow this difference to bear out in player performance rather than as a consequence of service and reception. Furthermore, the reception period should only be present for one possession; all possessions afterwards are classified as 'transition'. Thus, applying a blanket 'serving' team adjustment is not necessarily accurate - especially for the hit-based metrics.

5.4 Touch Plus-Minus Models (TPM)

A noteworthy element of volleyball is the ability to track every touch of the game. Fellingham utilises this and introduced the concept of a touch-based model to evaluate impact [1]. For the present paper, a regularised logistic regression was used as previously discussed. This was chosen for its simplicity when exploring the effect of these modifications.

5.4.1 Opponent-Presence (OPP)

Motivation

Regression based techniques generally model both teams. This not only serves to increase the quantity of information available, but helps adjust for the strength of opponent a player faces. Else, a player who only played against weaker teams may receive an inflated score. A potential weakness of Fellingham’s implementation is that the opposition was not included in the regression. Whilst this divorces itself from a purely touch-based model, a highly important source of information is provided.

Implementation

Court-presence of the opponent was included. The presence-based score should be generated independent of the touch-based score, otherwise the model will attempt to equalise these values. This is a relatively simple modification, simply including the identity of the players on the opponent’s team with the OPP prefix. The updated segments block for touches is given in Table 5.4

Table 5.4: An example excerpt from the touch segments including the opposition

Par1	Par2	Par3	Par4	Opp1	...	Opp6	Value
Aaron				OPP Bella	...	OPP Bobby	0
Bobby	Bella	Bismack		OPP Aaron	...	OPP Astarion	0
Anthony	Astarion	Angus	Alex	OPP Bella	...	OPP Bobby	1
Bobby				OPP Aaron	...	OPP Astarion	-1

Expectations

Anticipated to significantly improve model prediction performance. The value for the OPP presence-score will show positive correlation with the presence-based values. Furthermore, this will correlate more strongly with the unadjusted presence-based models, as a serving adjustment has not been implemented.

5.4.2 Skill-Touch Plus-Minus (STPM)

Motivation

As an extension upon the base skill-touch model, it is feasible to attach the skill being performed to each player touch. This should give the model far more context when assessing player touches. Of particular note is the inclusion of the skill of serving - which has demonstrated significant influence on the presence-based model.

The skill indicator provides substantial information and opportunities, and will be used as the base subject for most modifications in analysis.

Implementation

Implementation of the skill-touch has been covered prior, with example of modified skill-touches given in Table 4.4.

Limitations

Each skill applies a different tag to these players, roughly multiplying the number of predictors by 6. This can cause data sparsity issues, especially in smaller datasets.

Expectations

Model accuracy and predictive power will increase. However, this may be dependent on sample sizes. In terms of average score for positive outcomes, the list in descending order will be: Attack, set, freeball, block, reception, serve, dig. This is informed by domain knowledge; specifically how often a skill appears in isolation, and its ability to score a point.

5.4.3 Previous Action (prevact)

A purely touch-based approach for including the opposition may be through including the previous action. An extremely prominent aspect of volleyball is that each action is heavily influenced by the prior action [30]. For example, digging a poor attack attempt will generally have a better outcome than a dig from a powerful spike.

This approach should give the influence of a particular action on the opponent's subsequent ability to get a point or make an error. Using the base model, a serve is evaluated based only upon its immediate point-scoring or point-losing ability. However, the purpose of a serve is typically to disrupt the opponent's chance of forming an in-system attack.

Implementation

For the player matrix, the final touch of the previous possession is simply included in the subsequent possession. An example is provided in Table 5.5.

Table 5.5: An example of touch-based segments including the previous opponent action

Par1	Par2	Par3	Par4	PrevOpp1	Y
Aaron Serve					0
Bobby Reception	Bridget Set	Banksy Attack		Aaron Serve	0
Alex Block	Angel Dig	Anthony Set	Adrian Attack	Banksy Attack	1
Bobby Dig				Adrian Attack	-1

In the construction of the player matrix - the opponent's touch is indicated with a -1. The matrix construction is given in 5.6.

Table 5.6: An example of the player matrix when including the previous action.

Aaron Serve	Bobby Reception	Bridget Set	Banksy Attack	Alex Block	Angel Dig	Anthony Set	Adrian Attack	Bobby Dig	Y
1	0	0	0	0	0	0	0	0	0
-1	1	1	1	0	0	0	0	0	0
0	0	0	-1	1	1	1	1	0	1
0	0	0	0	0	0	0	-1	1	-1

Limitations

The final touch in a possession affects both the current outcome for the player's team, and the subsequent outcome for the opposing team. In the current methodology, the recorded skill-touch is duplicated, causing the regression model to treat these two effects as equivalent. However, this equivalence is not inherently valid. For example, consider a player's serve: An extremely passive server will rarely make errors (high rating in terms of avoiding negative outcome), but will similarly not induce an error in the opponent (low rating in terms of causing negative outcome). In the present framework, the models treat these effects as identical, despite representing distinct outcomes.

Furthermore, it is only the prior action that is included. A perfectly in system attack (set by setter) will be treated identically to an out of system scramble attack (set by non-setter).

Expectations

Accuracy will see some improvement. Serving will see great differentiation between players as it is no longer purely the point-scoring ability of a serve that is assessed. Some of the issues from error conditions will be alleviated. For defensive actions, they will no longer be treated in isolation and thus have greater importance and variability between players.

5.4.4 Previous Sequence (prevseq)

Motivation

An extension of the **prevact** augmentations is to include the entire sequence of opponent touches. This may accentuate the effect.

Implementation

The implementation is very similar to the previously described **prevact** augmentations - except all touches from the opponent are included. These opponent touches are similarly logged with a -1 in the player matrix.

Table 5.7: An example of touch-based segments including the previous opponent action

Par1	Par2	Par3	Par4	PrevOpp1	PrevOpp2	PrevOpp3	PrevOpp4	Y
Aaron Serve								0
Bobby Reception	Bridget Set	Banksy Attack		Aaron Serve				0
Alex Block	Angel Dig	Anthony Set	Adrian Attack	Bobby Reception	Bridget Set	Banksy Attack		1
Bobby Dig				Alex Block	Angel Dig	Anthony Set	Bobby Reception	-1

The resulting player matrix for this example is given in 5.8.

Table 5.8: An example of the player matrix when including the previous action.

Aaron Serve	Bobby Reception	Bridget Set	Banksy Attack	Alex Block	Angel Dig	Anthony Set	Adrian Attack	Bobby Dig	Y
1	0	0	0	0	0	0	0	0	0
-1	1	1	1	0	0	0	0	0	0
0	-1	-1	-1	1	1	1	1	0	1
0	0	0	0	-1	-1	-1	-1	1	-1

5.4.5 Sum-To-One Normalisation (stone)

Motivation

A touch-based regression model suffers from a significant characteristic: The number of agents is dynamic. As a consequence of this, there may be observations where a single player is present; resulting in a very strong signal in the regression. These error conditions may teach the model a player's 'pure' effect comes from their errors.

Implementation

To accommodate for this, the design matrix is augmented to ensure each row adds up to one. This should place less value on the singular touch outcomes when multiple touches are present. An example of the resultant player matrix is given in Table 5.9. When including the previous sequence of touches, these too are normalised to -1.

Table 5.9: An example of the player matrix when touches are normalised to sum to one

Aaron Serve	Bobby Reception	Bridget Set	Banksy Attack	Alex Block	Angel Dig	Anthony Set	Adrian Attack	Bobby Dig	Y
1	0	0	0	0	0	0	0	0	0
0	1/3	1/3	1/3	0	0	0	0	0	0
0	0	0	0	1/4	1/4	1/4	1/4	0	1
0	0	0	0	0	0	0	0	1	0

Expectations

The value of single-touch skills (serves, blocks) will increase. Accuracy has the potential to improve.

5.4.6 Presence-Based Hitters (phitters)

Motivation

This augmentation is designed to regulate for any participation biases. By considering the hitters on court, rather than the specific hitter - a more correct value can be attributed to the players that support these hitters. Furthermore, this should include some presence-based effects. For example, a highly effective middle blocker should act as a threat, opening up the other positions.

Implementation

To more accurately model the dynamics on the court, the hitters can be recorded based on presence rather than by touch. In this model, whenever an attack is made, the available hitting players are recorded. These are the outsides, middle blocker, and opposite hitter.

Table 5.10: An example of touch-based segments utilising presence-based hitters

Par1	Par2	Par3	Par4	Hit1	Hit2	Hit3	Hit4	Value
Aaron Serve								0
Bobby Reception	Bridget Set			Banksy HIT	Bismack HIT	Blufus HIT	Badger HIT	0
Alex Block	Angel Dig	Anthony Set		Adrian HIT	Alex HIT	Antonio HIT	Arfus HIT	1
Bismack Dig								-1

Expectations

Accuracy will improve significantly. However, this improvement will capture other aspects of court-presence rather than pure effect on offensive possessions. The values of reception, sets, and digs will become shifted.

5.5 Interim Results for Skill Averages

We can observe some of the trends that have been applied by some of these augmentations. For the skill-touch model, we will find the averages of each of the skills, both from the base case and when including some previous touches. This is given in Table 5.11.

For these skills, we will observe their influence on positive outcomes. The trends follow our general expectations. In the base case, attacks and sets are by far the most conducive to winning a point. Fundamentally, the chances of scoring a point without these actions is very low. In descending order, the defensive touches are freeballs, reception, and digs. This trend follows the controllability and degree of difficulty of these actions.

Table 5.11: Average coefficients for skills from the base Skill-Touch Plus-Minus model

Skill	BASE ⁺	BASE ⁻	Prevact ⁺	Prevact ⁻	Prevseq ⁺	Prevseq ⁻
Serve	-0.734	-1.509	-0.796	-0.745	-1.001	-0.550
Reception	-0.615	0.484	-1.112	0.347	-1.351	0.518
Set	0.676	-0.983	0.958	-0.968	0.581	-0.864
Attack	1.901	-1.426	0.947	-1.294	1.135	-1.076
Block	0.592	0.041	0.923	-0.439	0.408	-0.344
Dig	-1.056	0.681	-0.135	0.259	-0.768	0.376
Freeball	-0.259	-0.054	-0.155	0.267	-0.877	0.427
Intercept	-1.953	0.050	-1.600	-0.680	-1.355	-0.939

Note: BASE⁺ refers to the coefficient for POSITIVE outcomes for the basic STPM model.

By including the previous action, several shifts in the coefficients occur. Reception becomes the biggest inhibitor for positive outcomes. Likely because the serve features a negative score - and all serves are met by reception. Hence almost all observations will feature a (Reception - Serve) term. Attacks become significantly less absolute in terms of points winning - dropping from a 1.90 rating to 0.95. The defensive actions of blocks and digs also increase significantly - where digs are no longer viewed as extreme positive inhibitors (from -1.1 to -0.14). This is likely due to the (Attack - Dig) or (Attack - Block) observations being increasingly common.

For the case of the entire previous sequence, these values shift. It is likely conflation effects will arise as actions such as reception will receive scores for their influence on the opponent. The influence a skill has tends to diminish as the play continues [30].

Chapter 6

Model Evaluation Methodology

Each of the prior techniques returns a regression coefficient for each player and their tag. The linear regressions indicate a player's influence on the outcome for the relevant observation. For points, this is per rally, and for hits, this is per hitting attempt. In the logistic case, the regression returns two predictor values for each player. These are the log-odds indicator of each player's influence towards a positive or negative outcome. Whilst this grants some insights, further computation is required to convert this to an actionable value. A 'points added' value will be computed through each metric to directly compare across models.

For the following sections, we will use the basic Skill-Touch Plus-Minus (STPM) as the default model of reference for the skill-touch models.

6.1 Calculating Points Added

6.1.1 Relative Performance

Each player's score is evaluated relative to the average level of play. The baseline for 'average' is evaluated in two ways: Relative to the role; and where applicable, relative to the skill. For example, the weighted average for attacks from the middle is (+2.14), whereas the weighted average for all attacks is (+1.90). It is generally in these logistic cases where values are non-zero and vary significantly by position.

The relative coefficient for a player is simply the difference between a player's predictor and the weighted average. It is only for the skill-touch models where skill-relative values can be taken. The computation for a player's relative score is given in (6.1).

$$r_{i,\tau,y} = \beta_{i,\tau,y} - \frac{\sum_{k=1}^n \beta_{k,\tau} \omega_{k,\tau}}{\sum_{k=1}^n \omega_{k,\tau}} \quad (6.1)$$

Where $r_{i,\tau,y}$ refers to the relative coefficient of player i in context τ in regards to outcome

y (positive or negative). This situation term τ represents the specificity of the skill. For example, it can represent the context of role-relative **Outside|Attack**, or the context of skill-relative **Attack**. Here, $\omega_{k,\tau}$ is the weighting term and denotes the number of appearance. Hence, $\beta_{k,\tau,y}$ and $\omega_{k,\tau}$ refers to the regression coefficient and weights for player k in context τ (for positive/negative outcome y). They are used to compute the weighted average.

6.1.2 Points Added Per Observation

The relative coefficient $r_{i,\tau,y}$ is used to calculate points added per observation. This value will be multiplied by the number of observations to form the final number of points added. For the linear models, the value already matches this case. E.g. For the point-based RAPM scores, a player with a relative score of +0.01 who has participated in 2500 possessions will have contributed +25 points compared to the average.

Logistic models require some additional computation. In the pursuit of identifying 'change in outcome probability' a baseline is needed from which the change in log-odds is sourced from. For example, digs are very likely to result in a negative outcome (25.5%) - and hence reducing negative outcomes is more valuable in this context than for the low-error of attacks (14.3%). These empirical probabilities are converted to log-odds for appropriate addition of the relative player coefficient. The formula used for calculating change in outcome probability is given in (6.2). The logit function $\frac{p}{1-p}$ is used to convert probabilities to log-odds.

$$\Delta p_{i,\tau,y} = \log\left(\frac{p_{\tau,y} + r_{i,\tau,y}}{1 - (p_{\tau,y} + r_{i,\tau,y})}\right) - \log\left(\frac{p_{\tau,y}}{1 - p_{\tau,y}}\right) \quad (6.2)$$

Here $\Delta p_{i,\tau,y}$ represents the change in probability for possession outcome y for player i in context τ . The newly introduced $p_{\tau,y}$ term refers to the overall probability of outcome y in context τ . For example, the previously mentioned 25.5% for digs to result in a negative outcome.

Role vs Skill Example

A role-relative approach will be used as the standard implementation. An example will be given for a middle attack (59.5% chance of scoring a point). The logit for this is taken (+0.384). The player has a +2.20 coefficient, compared to +2.14 for middles in general. This is a +0.06 relative score. Adding this to the logit results in 0.444, corresponding to a probability of 60.9%. Thus, the player produced +0.014 points.

For the skill-relative, an attack in general may have a 50.2% chance of scoring a point (logit of roughly +0). The weighted average of the attack coefficient is +1.90, and hence the player has a +0.3 relative coefficient. The new probability for scoring a point is 57.4%, and hence the player produced +0.074 points - a notably larger effect.

For presence-based circumstances, a baseline of 50% is taken in all cases. I.e. DEF and HIT.

6.2 Evaluation Metrics

6.2.1 Comparison with Team Points

A simple yet effective method of evaluating metric accuracy is to compare actual team performance with the computed values. This method is particularly valuable for comparing different techniques, as the outcome of 'points' is directly comparable. Because each player has an associated 'points added' value, the total performance for the team can be evaluated. This can then be compared with the actual team performance.

These metrics have been computed 'relative to average', and as such team performance should similarly be evaluated 'relative to average'. Using the raw value of how much teams outscored their opponents is incompatible, as it is possible a team faced weak opponents multiple times which would artificially inflate their performance.

A value referred to as Simple Rating System (SRS) will be computed. A regression is run comparing the teams playing and point each set. The coefficients returned by this regression indicate how many points a team will win or lose by against an 'average' team. This is the SRS score. The model for SRS is estimated using a ridge-penalised regression analogous to the RAPM framework, as given in (6.3). Furthermore, the ridge penalty has been applied from Equation (3.2). This mirrors the RAPM framework, where instead team margin of victory is modelled as the difference in strength between the home and visiting teams.

$$M_i = \alpha_0 + \alpha_i - \alpha_j + \varepsilon \quad (6.3)$$

Here, M_i refers to the margin of victory, α_0 to the average home-court effect, α_i and α_j refers to the SRS scores of home-team i and visiting-team j .

By multiplying a team's strength score α_i by the number of sets played, a 'total team points added' is computed. This indicates a total amount of points the team has produced compared to an average team.

To compare this to model expectations, the sum of 'points added' for each player on the team is calculated. By performing a regression on these two values, we can find the quality of fit. We will now compute this for the baseline RAPM performance.

```
model <- lm(srs_points_added ~ RAPM_raw_team_sum, data = team_success)
summary(model)$r.squared
```

This produces the R^2 metric for each model. From the R documentation, this is an

application of Equation (6.4).

$$R^2 = 1 - \frac{\sum_i R_i^2}{\sum_i (y_i - \bar{y})^2} \quad (6.4)$$

Here, R_i refers to the i -th residual. The y_i term refers to outcome i and $\bar{y}$ to the average. Hence $\sum_i (y_i - \bar{y})^2$ refers to the total variance of team points. An R^2 score near 1 indicates the model can explain the majority of team scoring and success.

For demonstration purposes, the results for RAPM are given in Figure 6.1 and the results for STPM given in Figure 6.2.

Limitations

Whilst this system is very useful for overall prediction accuracy, it does not necessarily bear correct rankings between players on a team. It may be that the impact of liberos is generally absorbed by the setter of vice versa.

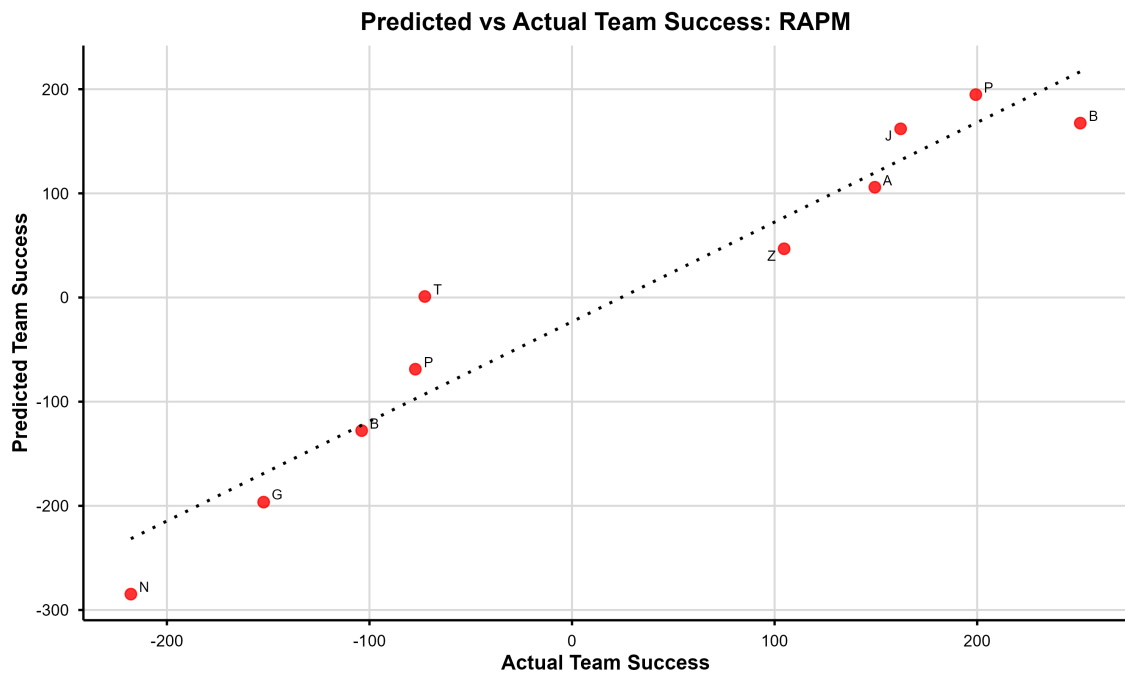


Figure 6.1: Regularised Adjusted Plus-Minus model predictions of team success

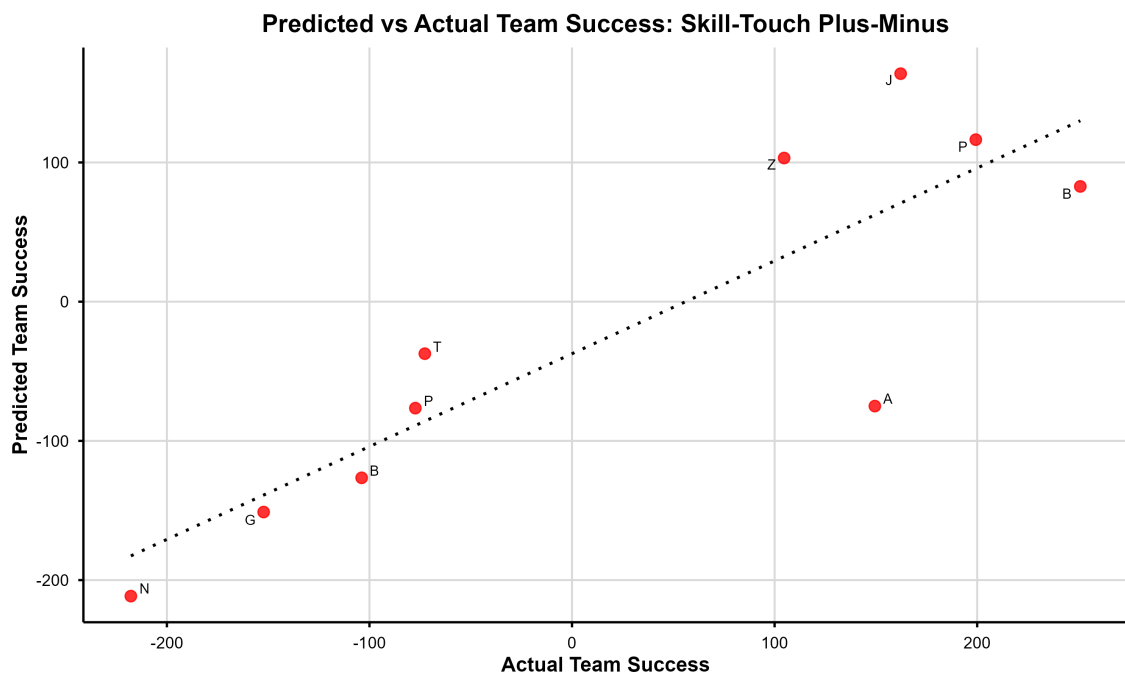


Figure 6.2: Skill-Touch Plus-Minus model predictions of team success

6.2.2 Cross-Validation Techniques

Cross-Validation R^2

Another powerful technique for determining accuracy is through cross validation. Model performance was assessed via 10-fold cross-validation, using the optimised λ for each regularised model. The models are then trained on 90% of the data, and predict the final 10%. This is repeated for each unknown 10% of data to produce a prediction matrix which can be compared with the recorded outcomes.

From the prediction matrix, many techniques can be used. The R^2 metric is used for this purpose once again, as it is a flexible and interpretable method for assessing model precision. The value represents the proportion of total variance explained.

Calibration Curves

An extension of this topic is calibration curves. Theoretically, this aims to compare the values in the prediction matrix with the actual outcomes by the use of 'bins'. In this case, 1% percentile bins were used. For example all prediction matrix values of 65% to 66% are binned together. The corresponding outcome of these predictions is also taken. A successfully calibrated model would find that for all predictions in the 65-66% bin, the outcome occurs $\sim 65.5\%$ of the time. In this way, the predicted and actual probabilities can be compared directly.

Here, R^2 will again be computed for these calibration curves. This value assesses the predictive power of these models. This should give an indication of the overall accuracy of these models, and how much confidence we as analysts can place in them.

An example of a calibration curve for Serving-Adjusted RAPM is given in Figure 6.3. Note the bimodal nature depending on which team is serving. The horizontal axis gives the models predicted rate of occurrence. The vertical axis gives the actual frequency of that bin. A perfect model would be a straight line, as all predictions are successful. A calibration curve for STPM is given in Figure 6.4.

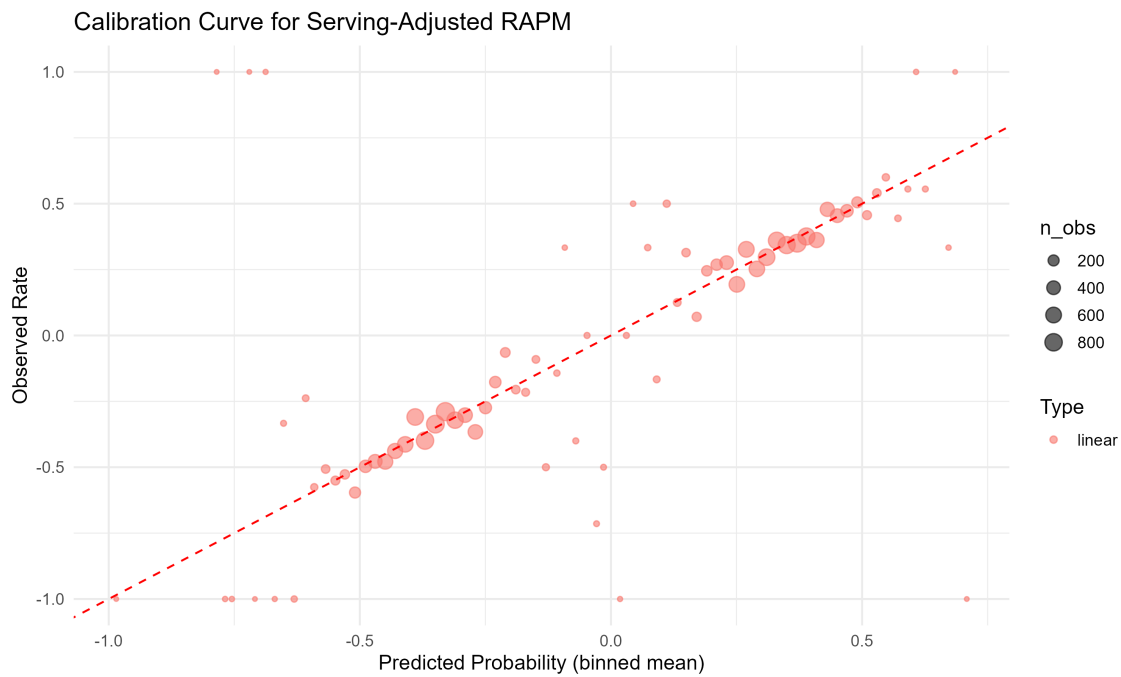


Figure 6.3: Calibration curve for Serving-Adjusted RAPM

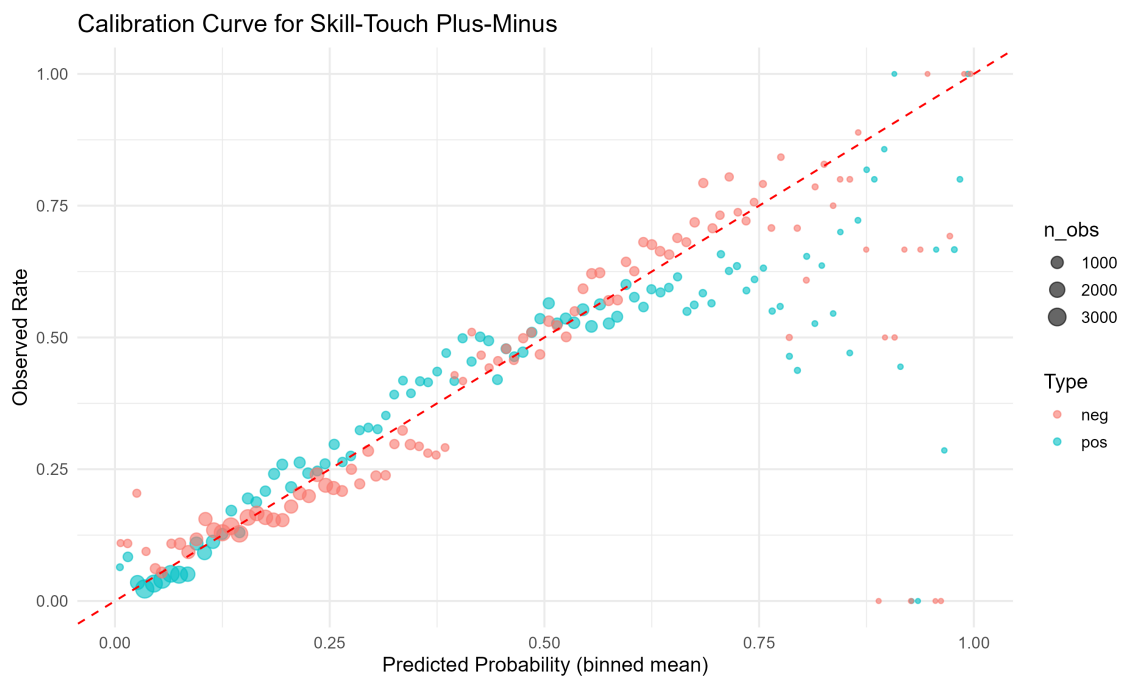


Figure 6.4: Calibration curve for Skill-Touch Plus-Minus

6.2.3 Model Biasing

There is no clear quantitative method for determining biases of the model. Instead, comparisons of the 'points added' across different categories will be made. The 'points added' metric is chosen as it is directly comparable across model, and furthermore has less contribution from the high variance impacts of low-involvement players. Players with just one observation exist, and their regression coefficients exhibit extreme regression values (e.g. +5 or -5, compared to typical values of +0.5 to +1.5).

Each model's allocation of explained points was examined by computing the proportion of absolute 'points added' for each role. This provides insight into which positions the models finds most influential. This process is also computed for each skill in those that feature them. For example, if serves account for 1,500 points added out of 10,000 points total, then 15% of points are determined by the differences between players in the serve. This proportion breakdown will provide an indication of how each model modification shifts the relative weighting of each skill or position.

Another examination for model biases will be between direct correlation of player points added between model variances. For example, comparing the scores between raw and opponent-adjusted models. A high correlation would indicate that including opponent-presence does not affect the models valuations of a players points added through their touches. A similar comparison will be made between the role-relative and skill-relative models, indicating which skills are most sensitive to the changes in the setpoint of baseline performance.

These findings can be compared with those in prior literature to evaluate whether the observed relationships align with established trends in player impact modelling. Such comparisons may reveal whether the present models reproduce known biases or introduce new ones, thereby providing an additional layer of validation for the analytical framework.

6.2.4 Comparison with Best Players

Finally, the models will be compared with players receiving awards for this season. Whilst voting can be very contextual, a strong model should find these award winning players provide a strong positive impact for their team.

Serving-Adjusted RAPM will be used as a reasonable baseline to assess which players are most impactful. These players should improve their team's performance whilst they are playing. The top two performers in this model will also be used as reference.

These will be converted to a 'per 50 points' rate to normalise for a different amount of points played. Only players involved in at least 500 possessions will be considered in the rankings.

Chapter 7

Results

7.1 Comparison with Team Success

Figure 7.1 presents the normalised prediction of team success for all evaluated models.

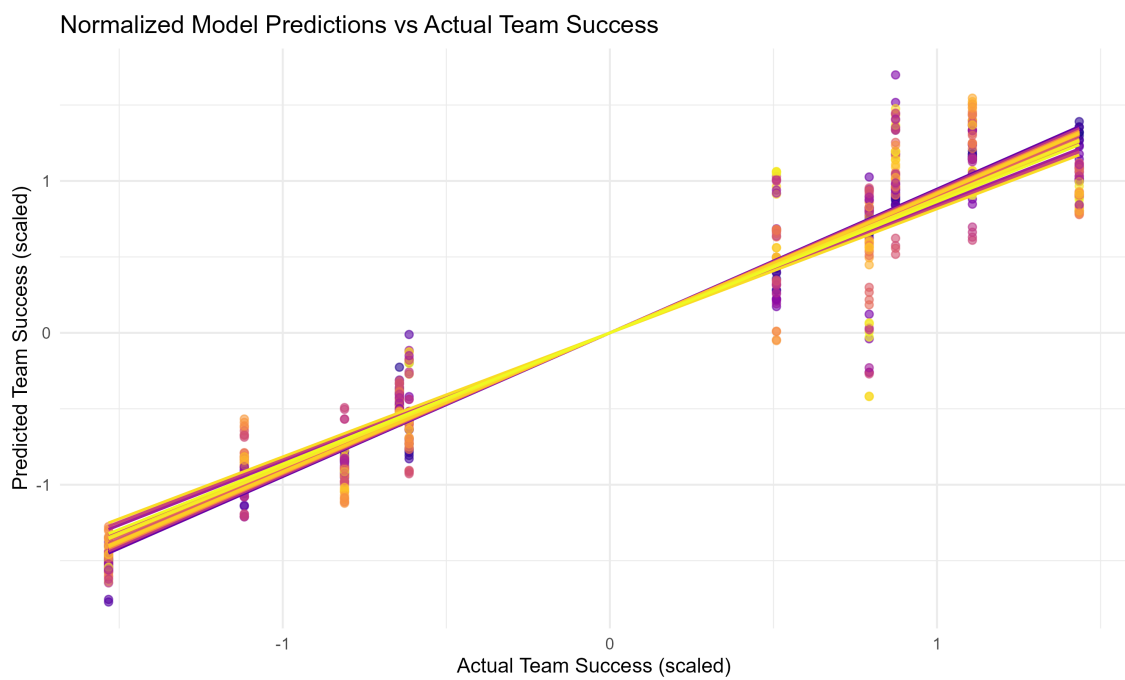


Figure 7.1: Plot of all models and their predictions of team success.

Table of Team Success Prediction Accuracy

The R^2 value comparing each model's predictions of team success to actual team success are provided in 7.1.

Table 7.1: The R^2 value of all models in relation to predicting team success

Base	Model	RB	RB FAB	SB	SB FAB
RAPM	Base	90.97	93.23		
	sadj	99.41	99.45		
	OD	98.49	98.63		
	OD_sadj	98.15	98.29		
	HitNet	97.22	97.88		
	HitNet_sadj	98.96	98.91		
TPM	Base	83.03	88.16		
	OPP	97.10	96.92		
STPM	Base	73.94	79.07	73.91	79.11
	OPP	95.84	96.07	95.69	95.9
	stone	82.62	85.72	82.63	86.88
	prevact	88.25	91.83	87.75	91.14
	prevact_OPP	91.44	92.67	91.8	93.11
	prevact_stone	95.17	95.44	94.48	95.47
	prevseq	92.33	93.25	92.59	93.41
	prevseq_def	86.28	86.78	86.42	86.93
	prevseq_stone	92.83	94.01	92.66	94.02
	phitters	76.59	83.66	76.53	83.27
	phitters_OPP	97.67	97.48	97.72	97.6

Note: RB = Role-Baseline. SB = Skill-Baseline. FAB = Front-And-Backcourt inclusive

Presence-based models tended to outperform those based on player touches. The most accurate metrics are purely presence-based. In this vein Serving-Adjusted RAPM (`RAPM_sadj`) is the strongest, with an R^2 metric of an impressive 99.28%. Trailing these are a cluster of other presence-based metrics.

The most accurate touch-based models is `STPM_phitters_OPP`, with a score of 96.27%. It features the most presence-based components of the touch-based models.

Touch-based models generally lagged behind. The base `STPM` metric is the worst performer at a mere 73.94% rating.

The methodology used to compute points added is slightly uncalibrated in terms of amplitude but can easily be scaled by comparing with actual team success.

7.2 Cross-Validation

The R^2 scores both for the raw prediction set and for the calibration curves are provided in Table 7.2. The calibration score gives the closeness of the predictions to the prediction bin.

Table 7.2: The R^2 value of all models in terms of predicting outcomes and closeness to calibration bins

Base	Model	Pred	Pred-FAB	Calib	Calib-FAB
RAPM	BASE	4.97	4.33	96.96	96.96
	sadj	13.14	13.17	98.57	98.19
	OD	0.33	0.44	99.24	99.50
	OD_sadj	0.32	0.42	99.18	99.26
	HitNet	2.48	2.37	98.15	98.26
	HitNet_sadj	7.12	7.17	98.22	98.59
TPM	BASE	12.18	13.88	94.27	92.06
	OPP	12.22	13.65	96.19	93.44
STPM	BASE	21.06	20.30	97.22	95.36
	OPP	20.94	19.98	97.34	95.15
	stone	21.21	20.67	97.09	96.18
	prevact	20.02	19.08	95.66	94.41
	prevact_OPP	19.86	18.79	95.81	94.61
	prevact_stone	18.60	17.69	95.00	94.40
	prevseq	19.59	18.64	96.91	94.71
	prevseq_OPP	19.40	18.31	96.79	95.18
	prevseq_stone	19.69	18.87	95.48	94.96
	phitters	20.94	20.28	96.60	94.58
	phitters_OPP	20.84	20.06	96.92	95.76

Note: Pred = R^2 score for cross-validation predictions. FAB = Front-And-Backcourt.
Calib = R^2 score for model predictions compared to calibration curves.

Touch-based models outperformed presence-based techniques in relation to cross-validation prediction. The STPM model holds a 21.06% cross-validation R^2 , compared to RAPM_sadj at 13.14%. However, this is largely due to the ability to predict much more aggressively. The prediction matrix of STPM has much steeper guesses. It has a much higher variance ($\sigma = 0.22$) compared to a stable metric such as RAPM_OD ($\sigma = 0.06$).

Using the calibration curves, all models performed effectively. However, the more stable presence-based metrics typically performed slightly more accurately.

7.3 Metric Biasing

Share of Points by Role

The percentage of total points accounted for by role is given in 7.3.

Table 7.3: Percentage of points explained by role for each model

Base	Model	Libero	Middle	Opposite	Outside	Setter
RAPM	BASE	22.4	19.2	19.5	15.4	23.6
	sadj	22.5	14.5	24.5	15.6	22.9
	OD	23	16.5	22.4	18.6	19.5
	OD_sadj	23.1	16.6	22.5	18.5	19.3
	HitNet	20.8	20.3	19.2	20	19.7
	HitNet_sadj	19.3	17.7	25	19.1	19
TPM	BASE	13	24.4	14.8	19.4	28.4
	OPP	21.9	17.8	19.4	25	15.9
STPM	BASE	10.4	17.5	13.9	18.9	39.4
	OPP	26	17.4	15.9	23.2	17.6
	stone	9.8	21.8	12.6	18.5	37.2
	prevact	15	17.5	13.3	19.1	35
	prevact_OPP	26.4	17	16.2	22.3	18.2
	prevact_stone	12.3	19.6	11.5	17.7	38.9
	prevseq	11.2	16.5	12.8	18.3	41.2
	prevseq_OPP	24.4	19.3	16.7	23.6	16
	prevseq_stone	10.6	19.6	13.1	19.3	37.5
	phitters	10.3	20.3	19.2	24.1	26.1
	phitters_OPP	24.3	19.2	19.1	22.5	14.8

Point contributions were somewhat evenly distributed across roles. With exception to the middle, each role accounts for the highest amount of points variance in at least one model. Touch-based models exhibit a clear bias towards the setter.

Share of Points by Skill

The proportion of total points accounted for by skill is given in 7.4. This excludes the points attributed to presence of the OPP models.

Interestingly, freeballs accounted for $\sim 4\%$ of total points, despite making up only 1.6% of touches.

Table 7.4: Proportion of points explained by skill for Skill-Touch Plus-Minus models

STPM Model	Serve	Recptn	Set	Atk	Block	Dig	Free	HIT
% of Touches	19.3	15.6	21.1	21.8	10.1	10.5	1.6	
BASE	13.4	13.5	23.9	20.4	13.3	11.9	3.7	
OPP	13.5	13.7	23.6	20.1	13.3	12.1	3.8	
stone	15.4	11.7	21.3	21.2	16	11.3	3.2	
prevact	9.7	17.3	21.1	18.1	15.3	14.1	4.3	
prevact_OPP	9.6	17.2	21.9	17.5	15.2	14.1	4.4	
prevact_stone	11.8	18.1	22.8	10.1	19	13.6	4.5	
prevseq	9.6	14.5	23.5	21	14.2	13.3	3.9	
prevseq_OPP	9.8	14.2	23.1	21.2	14.2	13.4	4	
prevseq_stone	10.2	13.8	21.9	22.5	16.1	12.2	3.4	
phitters	11.5	10.4	15.6		12.1	10	3.2	37.1
phitters_OPP	12.4	10.7	15.1		12.7	10.5	3.4	35.2
FAB_F		11.6	18.2	31.1	24.9	9.3	4.9	
FAB_B	24.2	16.5	26.3	11		18.5	3.5	
stone_FAB_F		10.2	15.6	31.7	30.2	8.2	4.1	
stone_FAB_B	30.4	14.6	24	11.6		16.1	3.3	
prevact_FAB_F		13	15.9	27.9	28.4	9.8	5	
prevact_FAB_B	18.9	19.3	23.5	12.1		21.9	4.3	
prevact_stone_FAB_F		13.6	18.3	16.8	36.5	9.4	5.4	
prevact_stone_FAB_B	21.8	20.2	26.2	6.8		20.6	4.4	
prevseq_FAB_F		12.8	17.2	31.3	25.2	8.9	4.7	
prevseq_FAB_B	18.1	17.2	26.7	12.3		21.9	3.9	
prevseq_stone_FAB_F		11.9	15.3	32.1	29.1	7.9	3.8	
prevseq_stone_FAB_B	20.7	17.7	23.9	13.2		20.9	3.6	
phitters_FAB_F		9.7	9.3		22.4	7.6	4.1	46.9
phitters_FAB_B	22	12.8	16.6			14.6	3	31
phitters_OPP_FAB_F		10.2	9.2		23	7.9	4.2	45.4
phitters_OPP_FAB_B	23.7	13.3	15.9			14.8	3.2	29.1

Note: HIT refers to the points attributed to presence-based hitting (**phitters**).

Correlations

The correlations of each augmented STPM model to the base STPM for each skill is given in Figure 7.2. The correlation heatmap for various presence-based components across these models is provided in Figure 7.3.

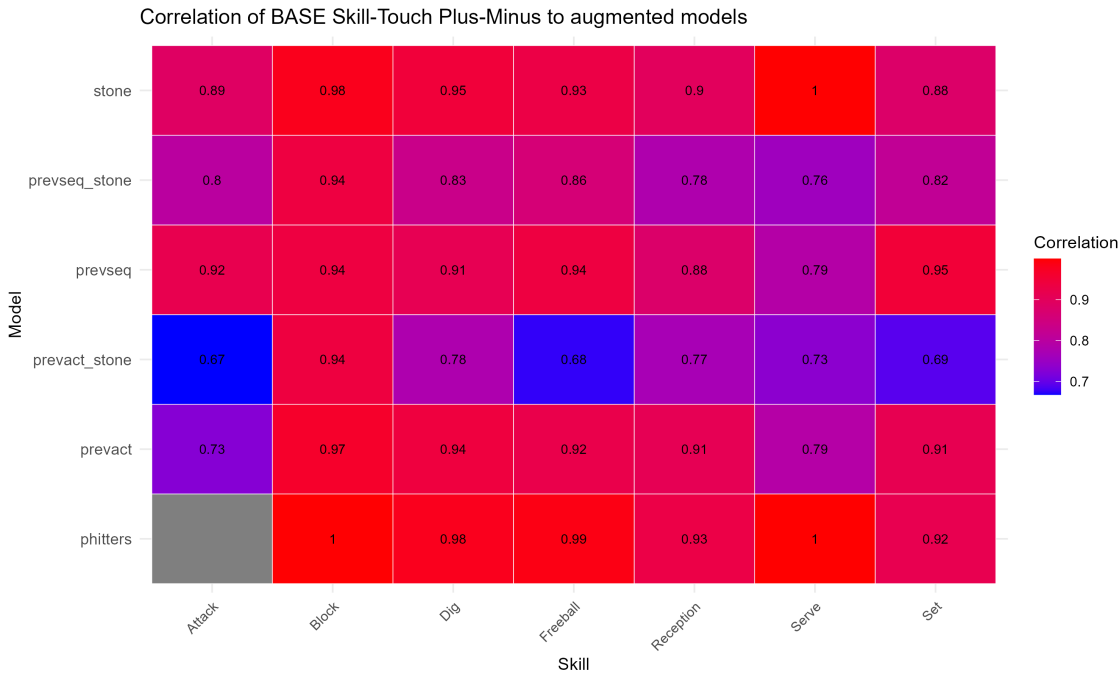


Figure 7.2: Correlations of player points for each skill for each augmented Skill-Touch Plus-Minus model

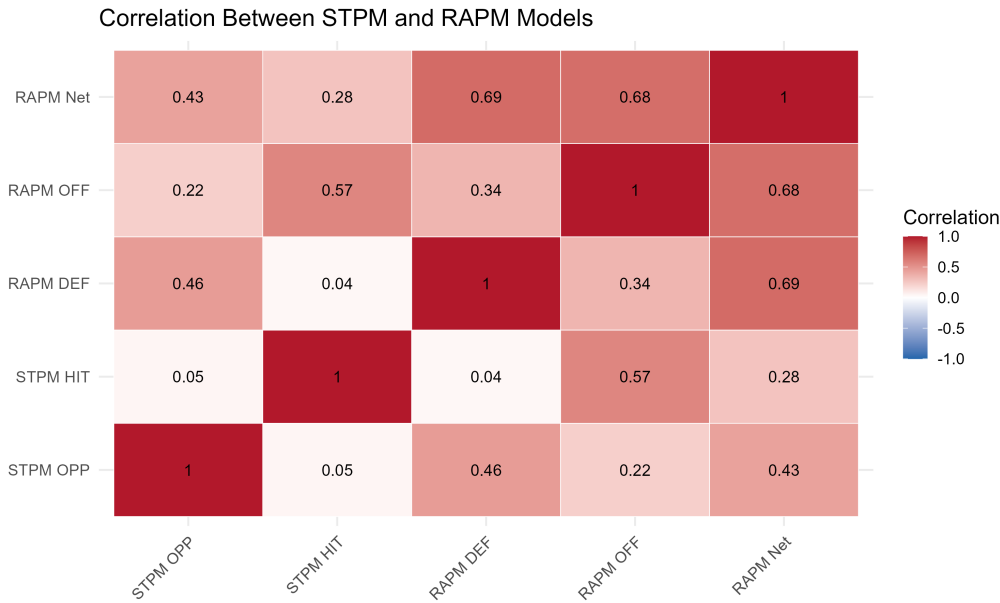


Figure 7.3: Correlations between presence-based aspects

7.4 Augmentations

This section evaluates the characteristics of each model augmentation relative to the baseline. A set of hypotheses and expectations have previously been made. Unless stated otherwise, comparisons for touch-based metrics are made using the role-relative baseline.

Generally, the player evaluation techniques explored within this paper have exhibited a strong explanatory power. This current paper aims to explore the key features and characteristics of these techniques.

7.4.1 Serving Adjustment

Applying a serving adjustment improved accuracy for presence-based models significantly. This is particularly notable for `RAPM_sadj`, which saw an increase in team-prediction accuracy from 90.97% to 99.41%. This is expected, as the serve is a massively important aspect of the game. Negligible change was observed for the hitting-split `OD` model however.

7.4.2 Front-and-Back Court Adjustment

Distinguishing between front-court and back-court generally improved results for these models - especially those using court-presence. Furthermore, it provides highly useful information for which substitutions may make the largest impact.

There are some other general trends that follow expectations. From Table 7.4, front-court metrics are dominated by attacking and blocking contributions (31.1% and 24.9% share). In the back-court, serving, setting, and digging (24.2%, 26.3%, and 18.5% share) became the most distinguishing aspects. This generally aligns with the order of importance scores found by other papers on the topics of skill importance [29].

7.4.3 Hitting-Based Observations

Using `RAPM_HitNet` mirrors the properties of the raw point-based `RAPM` very closely.

Splitting hitting into offensive and defensive components features much greater stability (`RAPM_OD`). This is most prominent in the calibration curves where these models dominate with a calibration R^2 of 99.24% (Table 7.2). Interestingly, the serving adjustment has little influence on results for this model. A possible explanation is that observations are made mid-rally - and thus serving players are on the receiving end of attacks much more often. A general presence-based model is agnostic to the nature of attacks.

However, these models do not particularly offer any new information nor use a much expanded dataset. The dataset features 17,000 points, yet only 19,000 hits. For leagues with longer rallies and less lethal attacks, these results may shift.

The offensive and defensive components exhibited nearly identical correlation with the overall `RAPM` score. In 7.3 there is a slight but significant correlation (+0.34) between the

components, indicating that there is some crossover between the two in volleyball.

7.4.4 Skill-Touches

Shifting to skill-touches (STPM) impaired predictions of team success significantly (from 83% to 74%). However, this may arise from miscalibration of the points added. There is clearly potential for a touch-skills model. In the cross-validation of Table 7.2, skill-touches have stronger predictive power (calibration R^2 of 95.5% vs 94%).

Otherwise, these models showed similar trends. Using skills allows for improved granularity, isolating where a player is strongest relative to their peers. This may be useful for separating between the attributes making strong and weak players. For example, any tendencies of effective passers, blockers, or servers.

In a redistributed segment of the dataset - and in most testing through this project - the use of skill-touches improved predictions of team success. Clearly, there is instability in the regression coefficients for players - even exposed to identical players. The first rendition of data (196 games) yielded the subsection of results found in Table 7.5. The STPM model records a 78.84% accuracy compared to 72.61% for TPM.

Table 7.5: Comparison of model predictions of team success R^2 values across a different sample

Model	RB	RB FAB	SB	SB FAB
TPM	72.61	71.69		
TPM_OPP	94.79	95.74		
STPM	78.84	81.84	75.5	78.36
STPM_OPP	97.17	97.15	97.04	96.9

7.4.5 Opposition Inclusion

Including the opposition significantly improved accuracy across almost all models. The STPM metric saw the biggest increase, from a team success prediction R^2 of 74% to 96% (+22%; Table 6.2).

Opposition-presence effects dominated the total explained variance, generally accounting for ~50% of total points added. Interestingly, this adjusted the point-share away from setters. From STPM to STPM_opp, the share drops from 39.4% to 17.6% when presence is included (Table 7.3).

However, these presence-based points are added almost completely independent to the touch-based component of the model. When correlating between the points gained for the Attack skill components of STPM and STPM_opp, the value is 0.9964, extraordinarily high. This correlation has been performed across each skill and each model. The full correlation

plot is given in Appendix A, but the averages have been summarised in Table 7.6. Including presence-based component has very little bearing on the values assigned to other actions.

Table 7.6: Average Skill Correlations — Base vs Opponent-Inclusive Models

Serve	Reception	Set	Attack	Block	Dig	Freeball	HIT
0.977	0.995	0.992	0.982	0.998	0.998	0.998	0.955

We compare the points attributed to opponent-presence value and various RAPM values in Table 7.7.

Table 7.7: Correlation between opponent-presence score and Regularised Adjusted Plus-Minus score

Model	Correlation
RAPM	0.013
RAPM_sadj	0.433

The **OPP** points have much stronger correlation with Serving-Adjusted RAPM, despite this adjustment never being applied to the **OPP** component. This may be from a similar phenomenon to the **RAPM_OD**, where the serving adjustment does not make a significant difference.

7.4.6 Presence-Based Hitters

Converting attack touches to a presence-based appearance made a small improvement on results in accuracy. Team-success predictions increased from 73.94% to 76.59% (Table 7.1).

From Figure 7.3, the **HIT** score exhibited significant correlation with the offensive aspect of **RAPM_OD** (0.564). This is unsurprising, as both models use attack observations. Additionally, it is nearly uncorrelated with the **OPP** score (0.05). This is promising from a perspective of splitting volleyball into offensive and defensive components

It was theorised that using presence-based hitters would help adjust for participation biases - primarily reception, digging, and setting. The resultant correlation between skills is given in Table 7.2. Whilst there is some credence to this (other skills at +0.99 correlation; setting and reception at ~0.92), the difference is marginal.

7.4.7 Previous Touches

Including the previous action (**STPM_prevact**) and previous sequence (**STPM_prevseq**) massively improved upon the base **STPM** model (+14 and +18 respectively). Performance was similar but generally diminished in cross-validation however. From Table 7.2,

`STPM_prevact_stone` is the second worst performer in terms of cross-validation (calibration R^2 of 95.00%) - yet is the best touch-based performer for a purely touch-based model (team success R^2 of 95.17%).

This modification was intended to capture aspects of the opposition. Consequently they did not work well in combination with a presence-based inclusion. In fact, the `STPM_prevseq OPP` combination is the only model that clearly suffered from including the opposition (team success R^2 drops from 92.33% to 86.28% from `STPM_prevseq` to `STPM_prevseq OPP`).

Serving was considered to be an aspect that is modelled very poorly under a base STPM implementation. However, serving became even less of a differentiator (from 15% to 10%; Table 7.4). A theory for this is that in a pure STPM model, players are differentiated by their pure point-scoring ability. As a consequence, the difference between aggressive or passive styles for serving was more prominent (I.e. Topspin serves and float-serves). From the `STPM_prevact` model, it may be that serves are ultimately relatively similar when accounting for impact on opposition. This aligns well with the findings of Liro & Gonzalez [27].

From Figure 7.2, this family of augmentations made the biggest change to the values of the STPM model. Offensive actions such as attacking and serving changed most significantly, whilst blocking was least affected.

7.4.8 Sum-to-One Normalisation

Whilst normalising did make significant improvements to accuracy in regard to team success (from a team success R^2 of 73.94% to 82.62%), there was little to no difference in terms of cross-validation. Changes in terms of overall role and skill share was minor.

However, when combined with the previous touches - particular with the previous action (`STPM_prevact_stone`) - significant differences in skill rankings were measured. From the correlation with the base model in Figure 7.2, the areas of Attack, Dig, Reception, Serve, and Set saw significant changes. The correlations dropped to roughly 0.7. This is the largest differences observed from any STPM model.

7.4.9 Role vs Skill Baseline

The difference between using the role or skill as a baseline is negligible.

A role-by-role correlation of the points for each skill in a role-baseline and a skill-baseline is given in Table 7.8.

Table 7.8: Correlation between role-based and skill-based points

Role	Serve	Reception	Set	Attack	Block	Dig	Freeball
Libero		0.990	0.670			0.981	0.952
Middle	1.000	0.996	0.754	0.870	0.993	0.976	0.992
Opposite	0.962	0.982	0.794	0.984	0.997	0.967	0.966
Outside	0.995	1.000	0.638	0.980	0.998	0.992	0.964
Setter	0.854	0.981	0.976	0.902	0.837	0.986	0.995

When correlating the metrics using the role or skill baseline, it is evident that they are very similar (mean $r > 0.9$). Not unexpectedly, the points generated by setting for non-setters shifts drastically. For attacking, the values for the high efficiency positions of middle and setter change most drastically (roughly 0.9 correlation).

Interestingly, the setter has a significantly different serving signal when taken on a skill relative basis. We can test whether they have any differences in their serving patterns.

role	`TPM_skills_pos Serve_avg`	`TPM_skills_neg Serve_avg`
1 libero	NaN	NaN
2 middle	-1.05	-1.60
3 opposite	-0.334	-1.15
4 outside	-0.462	-1.34
5 setter	-1.18	-2.10

Clearly, it bears out that setters tend to serve much more conservatively. They are both the least likely to ace and the least likely to error.

This change shifts the model in roughly the expected fashion - albeit to a fairly small effect.

7.5 Evaluation of Best Players

The players winning awards and their scores are given in Table 7.9.

Table 7.9: RAPM, TPM, and STPM (with previous action and sum-to-one normalisation) scores for award winners

Player	Team	Role	RAPM	STPM	PAS	R.%ile	S.%ile	PAS.%ile
Jerry	J	middle	0.33	0.87	0.78	80.77	84.62	92.31
Alucard	A	libero	0.54	-0.11	0.06	81.82	54.55	54.55
Alex	A	opposite	0.70	-0.63	0.44	82.35	29.41	94.12
Justin	J	outside	0.33	1.80	1.00	83.87	100.00	100.00
Anthony	A	setter	0.48	2.54	1.12	84.62	92.31	84.62
Peter	P	middle	0.75	0.12	0.75	96.15	65.38	88.46
Bismack	B	outside	0.88	0.56	0.23	96.77	74.19	70.97

Note: RAPM = Regularised Adjusted Plus-Minus, STPM = Skill-Touch Plus-Minus, PAS = STPM with Previous-Action inclusive with Sum-to-one normalisation. _%ile = Percentile

Percentiles were determined based on those who played in at least 500 points. Award winners typically had high RAPM scores, and all award winners are within the top 20% for their position. Furthermore, the MVP was 6th overall in the RAPM rankings.

Touch-models exhibited mostly positive scores. Justin and Antonio especially are considered very strong players in this regard. However, some players such as Alex are considered not only below neutral, but some of the worst players at their position.

We shift our attention to the players ranked 1st and 2nd in `RAPM_sadj` in Table 7.10.

Table 7.10: RAPM, TPM, and STPM (with previous action and sum-to-one normalisation) scores for RAPM leaders

Player	Team	Role	RAPM	STPM	PAS	R Rank	S.%ile	PAS.%ile
Jerry	J	libero	1.29	1.10	0.81	1	100.00	100.00
Angel	A	middle	0.85	-0.67	-0.18	1	26.92	46.15
Zion	Z	opposite	1.54	0.62	0.33	1	82.35	82.35
Perry	P	outside	1.22	-0.51	-0.08	1	35.48	48.39
Jesse	J	setter	0.74	-2.49	0.00	1	23.08	46.15
Zagreus	Z	libero	0.61	-0.01	0.42	2	63.64	90.91
Peter	P	middle	0.75	0.12	0.75	2	65.38	88.46
Patricia	P	opposite	0.98	0.19	0.13	2	64.71	58.82
Bismack	B	outside	0.88	0.56	0.23	2	74.19	70.97
Beatrice	B	setter	0.67	1.04	1.88	2	76.92	100.00

Note: RAPM = Regularised Adjusted Plus-Minus, STPM = Skill-Touch Plus-Minus, PAS = STPM with Previous-Action inclusive with Sum-to-one normalisation. %ile = Percentile

Those with strong RAPM scores should generally be improving their team’s performance whilst on the court. According to the touch-based models however, some of these players are extreme negatives. As an example Jesse is considered as one of the worst players in the dataset with a staggering valuation of -2.5 points per match-set.

When using the `STPM_prevact_stone` model, the results become slightly more consistent with Serving-Adjusted RAPM. However, it is evident that the set-points defined in `STPM` act as the baseline.

Chapter 8

Discussion

The player-evaluation methods developed in this study demonstrated significant capacity to predict match outcomes. The newly proposed skill-touch model relates the key skills to possession outcomes. A series of methodological augmentations were tested to assess their influence on model accuracy and interpretability.

8.0.1 Presence-Based Metrics

Presence-based metrics exhibited excellent calibration with match outcomes. Incorporating a serving adjustment proved essential for accurate modelling and ensuring a "zero average" baseline across all roles - particularly for the libero.

Detailing front-court and back-court positioning exhibited no degradation to prediction accuracy and provides actionable information in terms of substitution and lineups.

8.0.2 Applications

The `RAPM_sadj_FAB` model provides practical application for coaching decisions. For instance, the setter (hereafter "Henry") with the largest front-to-back-court disparity (minimum 500 possession) recorded a +0.67% effect in the front-court -0.95% in the back-court. By simply replacing Henry with an average setter in the back-court, this should improve the team's points per possession by +0.0095 per possession during these rotations. Over the course of a 50 possession game (and assuming equal front-court and back-court presence), this will contribute approximately +0.24 points per set. Using a simple regression of SRS compared to win percentage, this should improve the team's chances of winning by ~1.7%.

This example demonstrates the potential of these models to inform data-driven substitutions and lineup configurations.

8.0.3 Touch-Based Metrics

Touch-based models generally performed with a poorer accuracy. Despite this, there is clearly a strong predictive power to the technique in the manner of cross-validation.

These models present a host of implementation issues. They require significantly higher computational cost to run and are far more complex and sensitive to context. Difficulties include defining set-points for regularisation, specific adjustment for each role and skill, computing baselines for 'average' performance, and adjusting design matrices for non-binary outcomes.

Although the inclusion of opponent presence improved overall fit, it did not meaningfully adjust player touch-estimates. The resulting presence coefficient appeared largely orthogonal to baseline touch-values.

Shifting to a skill-touch methodology yielded mixed outcomes - higher predictive power but unstable accuracy. Theoretically, the improved data granularity offers diagnostic value for coaches, trainers, and in film analysis. The factors separating weak and strong performers in these categories can be assessed. However, in small datasets such as these, this introduces high dimensionality and sparsity.

In terms of adjusting distributions of player talent, the most influential augmentations were including previous touches, especially the directly prior action (prevact). Combining this with a sum-to-one normalisation produced the largest improvement in accuracy and the largest redistribution of player values.

Using presence-based hitters also made a significant change to player rankings. However, the change was not as large as expected - especially for reception, digging, and setting. The presence-based score captured some presence-based effects independent from the previously mentioned opposition-presence value.

8.1 Limitations

This dataset is very small - using a subset of a singular, relatively short season. Furthermore, it features an extremely high degree of multicollinearity - and oftentimes perfect collinearity. This can easily induce high sensitivity to these regression-based models.

Presence-based models are particularly vulnerable to these issues. On the other hand, the player matrices of touch-based models are inconsistent and endogenous.

In the presence-based model, liberos were assigned extremely positive values for the few possessions they are off for the serve. Similarly, a team's play style can massively influence these touch-based models. For example, the aggressiveness in which they operate for out-of-system play; which players set the ball when the need arises; blocking strategies. There is no common baseline across all teams.

All coefficients were computed relative to role or to the skill. However, this is a context-agnostic approach and does not take into account any strategy or coaching decision. For example, a player may be tasked with greater responsibility in reception, which leads to degraded individual scores but overall greater team performance.

8.2 Pillars of Sound Statistical Impact

Reliable player-impact modelling generally requires multiple seasons of data and relatively frequent player movement. In basketball, five-season windows (each with 82 games) are the norm for stable regression estimates.

The volleyball analytics sphere lacks a publicly accessible database of play-by-play information. In contrast, basketball holds an open-source, well-documented database of NBA games called Basketball Reference - an epitome of its kind. Without this data for volleyball, it is difficult to collaborate and perform studies upon a common dataset. It is an arduous task to even accurately recreate court-presence - let alone conduct the subsequent investigation.

Volleyball features many leagues and competitions all over the world. Unlike basketball analysis, which focuses primarily on the NBA, there is opportunity for multi-league evaluations of a player.

Ultimately, the sensitivity of these models is emblematic of the underdeveloped nature of volleyball stat-keeping. More so than evaluating player impact - these metrics reflect the structure and completeness of volleyball data.

Chapter 9

Conclusion

9.1 Summary of Work

Sports analytics has seen a limited application to volleyball. Producing a trustworthy metric for player evaluation will help push the sport forwards into a more developed state. This paper investigated a range of regression-based techniques and their properties. These approaches used presence, touches, and a new concept of skill-touches.

After a formidable data collection process, a set of functioning player matrices was constructed. A methodical set of augmentations were applied to determine their legitimacy in improving these player evaluation models. Several model evaluation techniques were developed to generate a comprehensive results list to compare and contrast the results from these models. Each approach was assessed on its ability to predict team success, prognosticate unseen data. Furthermore, a variety of correlations and point distribution comparisons were performed to test whether these changes had the intended effect. While certain hypotheses saw some empirical backing, others were not significantly substantiated in the results.

9.1.1 Properties of Regression as Applied to Volleyball

Whilst all of these models clearly express some predictive accuracy, there are many limitations. Like all professional sports, these players are extremely talented, and the difference between a strong and weak player is not necessarily an aspect we can gauge with high certainty. A difference in strategy or team-composition can alter these results to a highly significant extent.

Presence-based approaches continue to be effective and useful. Although, the limited size of the present data renders it improper for use. However, it is clear a serving adjustment and front-court and back-court differentiation are valuable options for analysis.

In the case of models based on touch - these suffer severely from the endogenous nature

of their observations. Despite this, including previous actions and applying normalisation have demonstrated a significant ability to shift these results and improve accuracy. Including presence-based elements has a useful, but isolated effect.

9.1.2 Data Considerations for Future Modelling

Multiple seasons of data and player movement is a necessity for this type of analysis. Clear situations for multicollinearity are embedded in the data.

In particular, touch-based models appear fundamentally constrained by the nature of their observations. Their sensitivity to context and event dependencies suggest that a restructuring of the model is required to meaningfully capture player impact.

Presence-based methods continue to be highly stable and predictive. However, the ability to correctly map which players make an impact is undefined. Future literature should focus on identifying the contexts in which RAPM-derived values are most valid and informative.

9.2 Future Works

Ultimately, using a touch-based model is highly dependent on the context of the team. A complete structural shift would need to occur. It is simply unsuitable to use regression methods in a source where observation elements are dependent on touch outcomes.

If pursuing touch-based regression as implemented, the following avenues are available:

- **Varying Penalty:** The penalty term can be weighted based upon appearances. This applies a harsher penalty to those with few touches.
- **Shrinkage Setpoint:** An initial pass (or techniques such as hierarchical Bayesian regression) can be used to find the mean of each position or skill. This will produce a tailored, non-zero shrinkage setpoint.
- **Linear Regression:** Linear regression may behave in more stable fashion, and better capture the overall effect rather than balancing positive and negative outcomes.

Some promising alternatives using a regression-based methodology include:

- **Statistical Plus-Minus:** Presence-based approaches have inconsistent applicability. Instead, training a regression only on teams and players with appropriate data (significant on and off samples) can produce reliable RAPM values. These may then be used to develop a Statistical Plus-Minus model, relating player statistics to impact. This can then be ported to players with high collinearity.
- **Full-Sequence Touch-Models:** Immediate success and failure states are obvious in volleyball. The design matrices can be filtered only to include sequences featuring at least 3 touches. This eliminates the highly sensitive error conditions. These situations can be modelled separately. The combined result of these should give a

much better indication of the effectiveness of a player both when they succeed and when they fail. For example, it is uncomplicated to separately compute a player's error rate when serving.

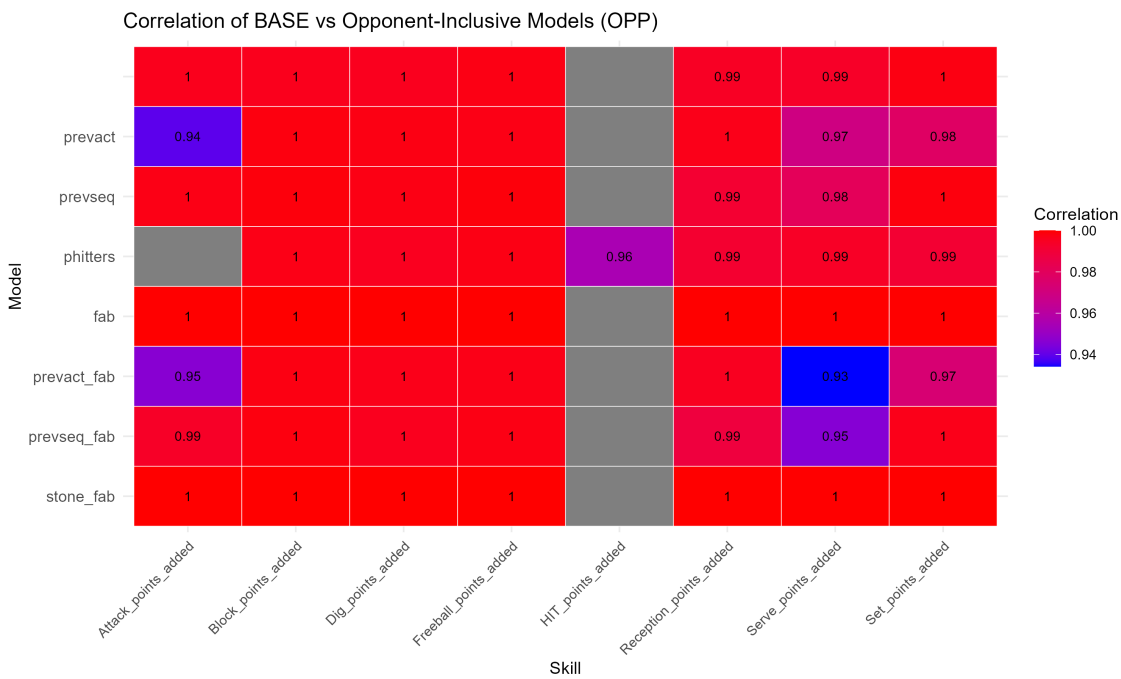
- **Machine Learning:** A methodology that is able to learn the relationships between the shapes and members of each sample has massive application. Including additional information should not detract from performance - but add. Other factors like setting distance and skill evaluations should be able to be incorporated without distracting from individual player performance.

A simple modification to the implemented presence-based RAPM is to simply include the identity of the server, rather than a blanket 'serving' element for both teams.

The concerns of this paper could be better assessed using a set of 'meta metrics' proposed by Franks [44]. This may particularly address the concern with RAPM in relation to sourcing values relative to the quality of teammates. By comparing RAPM across seasons - and especially in the case of a team trade - one can find how much player value fluctuates. Multi-year RAPM may be the strongest indicator, as team movement should globalise impact from team-relative to league-relative.

Appendix A

Full Correlation Plots



Appendix B

Match Scraping Code

```
1 import json
2 import re
3 import os
4 from bs4 import BeautifulSoup
5 import pandas as pd
6 import csv
7
8 def rotate_team_lineup(tl):
9     first = tl.pop(0)
10    tl.append(first)
11
12 def swap_players(players_list, ingoing, outgoing):
13     for i, player in enumerate(players_list):
14         if player["name"] == outgoing["name"] and player["number"] == outgoing["number"]:
15             players_list[i] = ingoing
16     return
17
18 def normalize_name(x: str) -> str:
19     if not x or not isinstance(x, str):
20         return ""
21     x = " ".join(x.strip().split())
22     # Split on anything that is not a Unicode letter (manual whitelist)
23     tokens = [t for t in tokens if t]
24     if not tokens:
25         return ""
26     first_initial = tokens[0][0]
27     last_name = tokens[-1]
28     return f"{first_initial}.{last_name}"
29
30 def setter_slot_from_rotation(rotation, positions_map):
31     for idx, p in enumerate(rotation):
32         pos = positions_map.get(p["name"], "")
33         if pos == "Setter":
34             return idx + 1
35     return "NA"
```



```

36
37 # Gets player positions.
38
39 PLAYER_POSITIONS = pd.read_csv("player_positions.csv", encoding="utf-8")
40 PLAYER_POSITIONS["player_id"] = PLAYER_POSITIONS["player_id"].astype(str)
41 ROLE_LOOKUP = dict(zip(PLAYER_POSITIONS["player_id"], PLAYER_POSITIONS["role"]))
42
43 # Start of main code.
44
45 def export_rallies_to_csv(rallies_with_context, game_info, player_positions, match_id,
46     filename="parsed_match.csv"):
47     data = []
48
49     for idx, rally in enumerate(rallies_with_context):
50
51         #print(f"Rally {idx}: actions = {len(rally.get('actions', []))}")
52
53         for jdx, touch in enumerate(rally["actions"]):
54             # Normalize and look up all 12 player IDs and roles
55             home_ids = [normalize_name(p["name"]) for p in rally["home_rotation"]]
56             visit_ids = [normalize_name(p["name"]) for p in rally["visiting_rotation"]]
57
58             home_roles = [ROLE_LOOKUP.get(pid, "unknown") for pid in home_ids]
59             visit_roles = [ROLE_LOOKUP.get(pid, "unknown") for pid in visit_ids]
60
61             home_setter_pos = next((i + 1 for i, r in enumerate(home_roles) if r ==
62 "setter"), "NA")
63             visiting_setter_pos = next((i + 1 for i, r in enumerate(visit_roles) if r ==
64 "setter"), "NA")
65
66             # Touch player's role
67             touch_pid = normalize_name(touch["player_name"])
68             touch_role = ROLE_LOOKUP.get(touch_pid, "unknown")
69
70             # TEMPORARY DEBUG SNIPPET
71             for i in range(6):
72                 if home_roles[i] == "libero" and i + 1 in [2, 3, 4]:
73                     print(f"Home libero in front row! Rally {idx}, Set {rally['set']}, "
74                         f"Rotation slot {i + 1}, Player = {home_ids[i]}, Team =
75 {game_info['home_team']}")
76                 if visit_roles[i] == "libero" and i + 1 in [2, 3, 4]:
77                     print(f"Visiting libero in front row! Rally {idx}, Set
78 {rally['set']}, "
79                         f"Rotation slot {i + 1}, Player = {visit_ids[i]}, Team =
80 {game_info['visiting_team']}")
81
82             data.append({
83                 "match_id": match_id,
84                 "point_id": idx,
85                 "time": "NA",

```

```

80         "team": touch["player_team"],
81         "player_number": touch["player_number"],
82         "player_id": touch_pid, # normalized
83         "skill": touch["skill"],
84         "evaluation": touch["evaluation"],
85         "timeout": "TRUE" if rally.get("timeouts") else "FALSE",
86         "substitution": "TRUE" if rally.get("subs") else "FALSE",
87         "point": "TRUE" if touch == rally["actions"][-1] else "FALSE",
88         "home_team_score": rally["home_points"],
89         "visiting_team_score": rally["visiting_points"],
90         "home_setter_position": home_setter_pos,
91         "visiting_setter_position": visiting_setter_pos,
92         **{f"home_p{i + 1}": rally['home_rotation'][i]['number'] for i in
range(6)},
93         **{f"visiting_p{i + 1}": rally['visiting_rotation'][i]['number'] for i in
range(6)},
94         **{f"home_player_id{i + 1}": home_ids[i] for i in range(6)},
95         **{f"visiting_player_id{i + 1}": visit_ids[i] for i in range(6)},
96         **{f"home_role{i + 1}": home_roles[i] for i in range(6)},
97         **{f"visiting_role{i + 1}": visit_roles[i] for i in range(6)},
98         "role": touch_role,
99         "set_number": rally["set"],
100        "team_touch_id": touch["team_touch_id"],
101        "home_team": game_info["home_team"],
102        "visiting_team": game_info["visiting_team"],
103        "point_won_by": rally["winner"],
104        "winning_attack": "TRUE" if (
105            touch["skill"] == "Attack" and jdx >= len(rally["actions"]) -
2 and touch["player_team"] ==
106            rally["winner"]) else "FALSE",
107        "serving_team": rally["serving_team"],
108        "home_score_start_of_point": int(rally["home_points"]) - 1 if
rally["winner"] == game_info[
109            "home_team"] else rally["home_points"],
110        "visiting_team_start_of_point": int(rally["visiting_points"]) - 1 if
rally["winner"] == game_info[
111            "visiting_team"] else rally["visiting_points"],
112    })
113
114
115
116
117    df = pd.DataFrame(data)
118    df.to_csv(filename, index=False, encoding="utf-8")
119
120
121    def process_all_matches():
122        with open("plusliga_metadata.json", "r", encoding="utf-8") as f:
123            metadata = json.load(f)
124

```

```

125 with open("player_positions.json", "r", encoding="utf-8") as f:
126     player_positions = json.load(f)
127
128
129 for entry in metadata:
130     match_id = entry["match_id"]
131     html_path = entry.get("html_path", f"rendered_matches/{match_id}_rendered.html")
132
133     if not os.path.exists(html_path):
134         print(f"Skipping {match_id}: HTML file not found at {html_path}")
135         continue
136
137     print(f"Parsing match {match_id} → {html_path}")
138
139     with open(html_path, "r", encoding="utf-8") as f:
140         soup = BeautifulSoup(f, "html.parser")
141
142         # Extract initial roster information : Home teams and players.
143
144         roster_rows = soup.find_all("vsw-roster-play-by-play")
145         lineup_rows = soup.find_all("vsw-lineup-play-by-play")
146
147         # =====
148         # ROSTER EXTRACTION | WHOLE GAME
149         # =====
150
151         home_team_roster = {}
152         visiting_team_roster = {}
153
154         for roster in roster_rows:
155             # Get all team divs (home and away) inside this roster
156             team_divs = roster.select("div.team") # gets all with class 'team'
157
158             for team_div in team_divs:
159                 # Get team name
160                 team_name_tag = team_div.select_one(".team-name .name")
161                 team_name = team_name_tag.text.strip() if team_name_tag else "Unknown
Team"
162
163                 # Get players
164                 players = []
165                 for player_div in team_div.select(".team-roster .player"):
166                     number = player_div.select_one(".player-nr")
167                     name = player_div.select_one(".player-name")
168                     if number and name:
169                         players.append({
170                             "number": number.text.strip(),
171                             "name": normalize_name(name.text.strip())
172                         })
173

```

```

174         # Check for 'right' class to decide home/away
175         if "right" in team_div.get("class", []):
176             visiting_team_roster = {
177                 "team_name": team_name,
178                 "players": players
179             }
180         else:
181             home_team_roster = {
182                 "team_name": team_name,
183                 "players": players
184             }
185
186         # =====
187         # LINEUP EXTRACTION / SET BY SET
188         # =====
189
190         lineups_by_set = []
191
192         # Each .events-container represents one set
193         set_containers = soup.select("div.events-container")
194
195         rallies_with_context = []
196         team_touch_id = -1
197
198         for container in reversed(set_containers): # Go through backwards for
199             chronological order.
200             set_number_tag = container.select_one(".set-bar .set-number") # Set number
201             text.
202             if not set_number_tag:
203                 continue # skip if not found
204             set_number_text = set_number_tag.text.strip()
205
206             # Extract the numeric set number
207             import re
208
209             match = re.search(r"\d+", set_number_text)
210             set_number = int(match.group()) if match else -1
211
212             # Now find the lineup block for this set
213             lineup = container.find("vsw-lineup-play-by-play")
214             if not lineup:
215                 continue # skip if lineup not found
216
217             # Extract home and away teams
218             right_team_lineup = {}
219             left_team_lineup = {}
220
221             team_divs = lineup.select("div.team")
222             for team_div in team_divs:
223                 team_name_tag = team_div.select_one(".team-name .name")

```

```

222         team_name = team_name_tag.text.strip() if team_name_tag else "Unknown
Team"
223
224     players = []
225     for player_div in team_div.select(".team-lineup .player"):
226         number = player_div.select_one(".player-nr")
227         name = player_div.select_one(".player-name")
228         if number and name:
229             players.append({
230                 "number": number.text.strip(),
231                 "name": normalize_name(name.text.strip())
232             })
233
234     if "right" in team_div.get("class", []):
235         right_team_lineup = {
236             "team_name": team_name,
237             "players": players
238         }
239     else:
240         left_team_lineup = {
241             "team_name": team_name,
242             "players": players
243         }
244
245     lineups_by_set.append({
246         "set_number": set_number,
247         "right_team": right_team_lineup,
248         "left_team": left_team_lineup
249     })
250
251     # =====
252     # RALLY EXTRACTION / RALLY BY RALLY
253     # =====
254
255     event_blocks = container.find_all("vsw-event-play-by-play")
256
257     # ROTATING ELEMENTS
258     =====
259
260     # Managing rotations.
261     rotations = {
262         "left_team": left_team_lineup,
263         "right_team": right_team_lineup,
264     }
265
266     coin_toss_items = container.select("div.coin-toss-item")
267     for item in coin_toss_items:
268         label = item.select_one("label")
269         if label and "Serving team:" in label.text:
270             serving_team = item.select_one("span.value").text.strip()

```

```

270 winner = "Left" if left_team_lineup["team_name"] == serving_team else "Right"
271
272
273 # VIDEO VERIFICATION STUFF =====
274
275 # Event Sequencing
276 event_types = [("Rally", "vsw-rally-play-by-play"),
277               ("Video", "vsw-video-challenge-play-by-play"),
278               ("Sub", "vsw-substitution-play-by-play"),
279               ("Timeout", "vsw-time-out-play-by-play"),
280               ("Libero", "vsw-libero-play-by-play"), ]
281 event_order = []
282
283 for event in reversed(event_blocks): # Goes through each event block and the
events within them.
284     found_types = [label for label, tag in event_types if event.find(tag)]
285     event_order.append(found_types)
286     seq_events = []
287     for i, item in enumerate(event_order):
288         for x in item:
289             seq_events.append((x, i))
290
291 rally_indices = [] # A bunch of tuples. Includes a mapping to the
appropriate video index.
292 for idx, tags in enumerate(event_order):
293     if "Rally" in tags:
294         rally_indices.append([idx, None]) # (rally_idx, video_idx or None)
295     if "Video" in tags:
296         rally_indices[-1][1] = idx
297 rally_video_map = {rally_idx: video_idx for rally_idx, video_idx in
rally_indices if
298                   video_idx is not None}
299
300 # OTHER RALLY ELEMENTS =====
301
302 left_points = "0" # Should probably be ints...
303 right_points = "0"
304
305 pending_subs = []
306 pending_timeouts = []
307 pending_liberos = []
308 pending_verifications = []
309
310 for i, event in enumerate(reversed(event_blocks)):
311     rally = event.find("vsw-rally-play-by-play")
312     sub = event.find("vsw-substitution-play-by-play")
313     timeout = event.find("vsw-time-out-play-by-play")
314     libero = event.find("vsw-libero-play-by-play")
315     video = event.find("vsw-video-challenge-play-by-play")
316

```

```

317         if rally:
318
319             # Gets rid of the 'rally repeated' things.
320             if rally.select_one(".rally-repeated"):
321                 continue
322
323             prevLeftPoints = left_points
324             prevRightPoints = right_points
325
326             # This gets the score.
327             if i in rally_video_map: # If there's a video verification, use
those scores
328                 video = event_blocks[len(event_blocks) - 1 -
rally_video_map[i]].find(
329                     "vsw-video-challenge-play-by-play")
330                 score_div = video.select_one(".score")
331                 left_points = score_div.select_one(".left").text.strip()
332                 right_points = score_div.select_one(".right").text.strip()
333
334                 video_info = video.select(".event-text-discription")
335
336                 pending_verifications.append({
337                     "team": video_info[0].text.strip(),
338                     "successful": video_info[1].text.strip()})
339             else:
340                 try:
341                     left_points = rally.select_one(".score .left").text.strip()
342                     right_points = rally.select_one(".score .right").text.strip()
343                 except:
344                     print(left_points, right_points, set_number)
345                     left_points = prevLeftPoints
346                     right_points = prevRightPoints
347
348                 home_points = left_points if home_team_roster["team_name"] ==
left_team_lineup[
349                     "team_name"] else right_points
350                 visiting_points = left_points if visiting_team_roster["team_name"] ==
left_team_lineup[
351                     "team_name"] else right_points
352
353                 prev_winner = winner
354                 winner = "Left" if int(left_points) > int(prevLeftPoints) else "Right"
355
356                 home_rotation = rotations["left_team"]["players"].copy() if
home_team_roster["team_name"] == \
357                 left_team_lineup["team_name"] else \
358                 rotations["right_team"]["players"].copy()
359                 visiting_rotation = rotations["left_team"]["players"].copy() if
visiting_team_roster["team_name"] == \

```

```

360     left_team_lineup[
361         "team_name"] else \
362         rotations["right_team"]["players"].copy()
363
364     rally_info = {
365         "home_points": home_points,
366         "visiting_points": visiting_points,
367         "winner": left_team_lineup["team_name"] if winner == "Left" else
right_team_lineup[
368         "team_name"],
369         "serving_team": left_team_lineup["team_name"] if prev_winner ==
"Left" else
370         right_team_lineup["team_name"],
371         "home_rotation": home_rotation,
372         "visiting_rotation": visiting_rotation,
373         "timeouts": pending_timeouts,
374         "subs": pending_subs,
375         "liberos": pending_liberos,
376         "verifications": pending_verifications,
377         "set": set_number,
378         "actions": [],
379     }
380
381     rallies_with_context.append(rally_info)
382
383     # Rotates the teams cause that happens AFTER the rally.
384     if prev_winner != winner:
385         rotate_team_lineup(
386             rotations["left_team"]["players"]) if winner == "Left" else
rotate_team_lineup(
387             rotations["right_team"]["players"])
388
389     # Clear pending events after attaching
390     pending_subs = []
391     pending_timeouts = []
392     pending_liberos = []
393     pending_verifications = []
394
395     # =====
396     # ACTION EXTRACTION
397     # =====
398
399     action_rows = rally.find_all("vsw-rally-playrow-play-by-play")
400     rally_actions = []
401
402     current_possession = None
403     for action in action_rows:
404         player_name =

```



```

405     normalize_name(action.select_one(".player-name").text)
406     skill = action.select_one(".skill").get_text(strip=True)
407     evaluation = action.select_one(".effect").get_text(strip=True)
408
409     # Find the child div that has the class "right" or "left"
410     alignment_div =
411     action.select_one("div.rally-playrow-play-by-play")
412     if "right" in alignment_div.get("class", []):
413         player_team = right_team_lineup["team_name"]
414     else:
415         player_team = left_team_lineup["team_name"]
416
417     if current_possession != player_team:
418         team_touch_id += 1
419     current_possession = player_team
420
421     player_number = "NA"
422     lineup_players = left_team_lineup["players"] if player_team ==
423     left_team_lineup[
424         "team_name"] else right_team_lineup["players"]
425
426     for p in lineup_players:
427         if p["name"] == player_name:
428             player_number = p["number"]
429             break
430
431     touch_actions = {"player_name": player_name, "player_number":
432     player_number, "skill": skill,
433     "evaluation": evaluation, "player_team":
434     player_team,
435     "team_touch_id": team_touch_id}
436     rally_actions.append(touch_actions)
437
438     rally_info["actions"] = rally_actions # Change in a sec.
439
440     # print(rally_info)
441
442     if libero:
443         player_in_block = libero.select_one(".player.in")
444         player_out_block = libero.select_one(".player.out")
445
446         player_in = {
447             "number": player_in_block.select_one(".number").text.strip(),
448             "name":
449             normalize_name(player_in_block.select_one(".name").text.strip()),
450             } if player_in_block else {}
451
452         player_out = {
453             "number": player_out_block.select_one(".number").text.strip(),
454             "name":

```

```

449     normalize_name(player_out_block.select_one(".name").text.strip()),
450     } if player_out_block else {}
451
452     pending_liberos.append({"in": player_in, "out": player_out})
453
454     if player_out and player_out.get("name") in [p["name"] for p in
left_team_lineup["players"]]:
455         t1 = left_team_lineup["players"]
456     else:
457         t1 = right_team_lineup["players"]
458
459     swap_players(t1, player_in, player_out)
460
461     elif sub:
462         # (same sub processing code as before)
463         icons = sub.select(".icon")
464         texts = sub.select(".event-text")
465
466         sub_events = []
467         for i, icon in enumerate(icons):
468             if i >= len(texts):
469                 continue
470             text_block = texts[i + 1]
471             number = text_block.select_one(".number")
472             first = text_block.select_one(".first")
473
474             player = {
475                 "number": number.text.strip() if number else "N/A",
476                 "name": normalize_name(first.text.strip()) if first else
"N/A",
477             }
478
479             if "out" in icon.get("class", []):
480                 player["direction"] = "out"
481             elif "in" in icon.get("class", []):
482                 player["direction"] = "in"
483             else:
484                 player["direction"] = "unknown"
485
486             sub_events.append(player)
487
488             player_in = next((p for p in sub_events if p["direction"] == "in"),
None)
489             player_out = next((p for p in sub_events if p["direction"] == "out"),
None)
490
491             pending_subs.append({"in": player_in, "out": player_out})
492
493             # Swapping subs.
494             if player_out and player_out.get("name") in [p["name"] for p in

```

```

left_team_lineup["players"]]:
494         t1 = left_team_lineup["players"]
495     else:
496         t1 = right_team_lineup["players"]
497
498         swap_players(t1, player_in, player_out)
499
500     elif timeout:
501         timeout_team = timeout.select_one(".event-text-discription")
502         pending_timeouts.append(timeout_team.text.strip() if timeout_team
else "Unknown")
503
504     game_info = {
505         "home_team": home_team_roster["team_name"],
506         "visiting_team": visiting_team_roster["team_name"],
507     }
508
509     export_rallies_to_csv(
510         rallies_with_context,
511         game_info,
512         player_positions,
513         match_id,
514         filename=f"parsed_matches/parsed_match_{match_id}.csv"
515     )
516
517     print(f"Finished match {match_id}")
518
519
520
521 if __name__ == "__main__":
522     process_all_matches()

```

Listing B.1: Code used to parse through raw HTML files from data scraping

Appendix C

Player Impact Code

The full list of R scripts is available on github at:

<https://github.com/NathanielGirdlerUON/VolleyballAnalyticsFYP>

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